

Article

Spatio-Temporal Feature Engineering and Selection-Based Flight Arrival Delay Prediction Using Deep Feedforward Regression Network

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Abstract: Flight delays continue to pose a substantial concern in the aviation sector, impacting both operational efficiency and passenger satisfaction. Existing systems, while attempting to predict delays, often lack accurate predictive capabilities due to poor modeling setups, insufficient feature engineering, and inadequate feature selection processes, leading to suboptimal predictions and ineffective decision-making. Precisely forecasting flight arrival delays is essential for improving airline scheduling and resource allocation. The aim of our research is to create a superior prediction model that surpasses current modeling approaches. This study aims to forecast airline arrival delays by examining data from five prominent U.S. states in 2023—California (CA), Texas (TX), Florida (FL), New York (NY), and Georgia (GA). Our proposed modeling approach involves feature engineering to identify significant variables, followed by a novel feature selection algorithm (CFS) designed to retain only the most relevant features. Delay forecasts were generated using our proposed Deep Feed Forward Regression Network (DFFRN), a five-layer deep learning approach designed to enhance predictive accuracy by incorporating extensively selected features. The findings indicate that the DFFRN model substantially outperformed conventional models documented in the literature. The DFFRN had the highest R^2 score (99.916%), indicating exceptional predictive efficacy, highlighting the efficacy of the DFFRN model for predicting flight delays and establishing it as a significant asset for improving decision-making and minimizing operational delays in the aviation sector.

Keywords: flight delay prediction; spatio-temporal data; deep feed forward regression network; feature engineering; air traffic management; regression analysis



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1. Introduction

Air travel is integral to the global economy, serving as a crucial enabler of international commerce, tourism, and personal connectivity. However, a persistent challenge facing this vital sector is flight delays, which have widespread repercussions. Flight delays not only frustrate passengers, leading to reduced satisfaction and potential loss of customer loyalty [1], but also impose considerable financial burdens on airlines. The U.S. Department of Transportation estimates that every year, aviation delays cost airlines billions of dollars, as well as frustrate passengers, leading to reduced satisfaction and potential loss of customer loyalty [1], and impacting customers due to missed connections and altered travel plans [2]. These include increased operational costs, compensation claims, and disruptions to scheduling and crew management [3]. Moreover, delays have cascading effects on airport

operations and the broader transportation network, causing logistical inefficiencies and diminishing overall service quality [4].

Several strategies are used to address flight delays, including schedule buffering, dynamic gate assignments, and air traffic management (ATM) techniques like rerouting and slot swapping. These methods aim to optimize airport operations and airspace usage to reduce congestion and delay impacts [5]. Resource reallocation strategies, such as aircraft or crew swaps, are also employed to minimize disruptions to subsequent flights [6]. However, the emergence of machine learning has transformed delay management by utilizing the extensive data generated from flight schedules, weather conditions, and air traffic records. The vast amount of data allows machine learning algorithms to predict delays with higher accuracy.

Accurately predicting flight delays offers significant benefits for airlines, passengers, airports, and air traffic controllers. For airlines, such predictions lead to optimized scheduling and resource allocation, reducing fuel consumption, crew management costs, and operational inefficiencies [7,8]. Passengers benefit from timely updates, allowing them to adjust travel plans with less stress, improving their experience and boosting customer satisfaction and loyalty [9]. Airports can better manage gates and runways, reducing congestion and enabling smoother operations [10]. Similarly, air traffic controllers can use accurate predictions to manage flight schedules more effectively, reducing bottlenecks and enhancing overall safety [11]. These improvements not only reduce operational costs but also enhance the overall efficiency of the flight ecosystem. Recent improvements in computational learning have shown promise in improving the accuracy of predictive models for flight delays. Specifically, deep learning methods like Long Short-Term Memory (LSTM) [12] and Convolutional Neural Networks (CNNs) [13] have been widely applied to extract complex patterns from large datasets.

The objective of this study is to advance the field of predicting aviation arrival delays by leveraging spatio-temporal feature engineering and a customized Deep Feed Forward Regression Network (DFFRN), a five-layer deep learning modeling approach designed to learn from spatio-temporal features. The dataset used in this study is sourced from Kaggle, specifically the “Flight Delay and Cancellation Dataset 2019–2023” (<https://www.kaggle.com/datasets/patrickzel/flight-delay-and-cancellation-dataset-2019-2023/data>, accessed on 6 December 2024), which was originally compiled and is provided by the U.S. Department of Transportation, Bureau of Transportation (<https://www.transtats.bts.gov>, accessed on 6 December 2024). The data cover a wide range of flight-related attributes, including flight delays, cancellations, and other operational data from 2019 to 2023. The dataset is structured as time-series data, with each record corresponding to an individual flight. Key variables include flight identifiers (e.g., flight number, carrier), scheduled departure and arrival times, actual departure and arrival times, delay information, cancellation, and diverted status. The time-series nature of the data allows for analysis of patterns and trends over time. To achieve our research objectives, we integrated spatial data (longitude and latitude) of different states. Feature engineering was used to derive meaningful variables, followed by a proposed correlation-based feature selection (CFS) algorithm designed to figure out the most relevant attributes for predictive modeling. The proposed DFFRN model, designed to effectively capture both linear and nonlinear relationships in the data, demonstrated superior performance compared to existing methods.

According to our research, the DFFRN model not only increases prediction reliability but also provides airport management with a useful tool for improving decision-making, allocating resources optimally, and eventually cutting down on operational delays. The key findings of this research are outlined below:

- We applied advanced feature engineering techniques to incorporate spatio-temporal information, significantly improving the model’s ability to predict flight delays.
- We introduced a novel feature selection method (CFS) that efficiently reduced data dimensionality, enhancing both computational efficiency and model performance.

- We created a five-layer Deep Feed Forward Regression Network (DFFRN), which outperformed the current state-of-the-art models.
- We examined flight delay patterns, including trends by state, departure–arrival delay relationships, seasonal variations, weekday vs. weekend patterns, and hourly and day–night trends.

The following is how this document is structured: Relevant works are presented in Section 2, which offers a thorough summary of current research in aviation delay prediction. The approach is explained in Section 3, including the feature engineering, modeling, and data processing procedures used in our investigation. The findings and discussion are presented in Section 4, where we evaluate the effectiveness of our suggested model and contrast it with baseline techniques. The work is finally concluded in Section 5, which provides a summary of our findings and recommendations for further research.

2. Related Work

Anticipating delays on flights accurately is essential for streamlining airport operations, lowering expenditures, and raising passenger satisfaction. Research predominantly leverages two approaches: classification, which categorizes flights as delayed or on-time, and regression, which provides quantitative delay predictions. Machine learning models like CNN, LSTM, and Random Forest have recently demonstrated significant potential in handling the complexities of flight delay prediction, especially in capturing spatial–temporal dependencies and enhancing predictive accuracy. This review details the application of these models in classification and regression studies.

2.1. Classification Approaches to Flight Delay Prediction

Classification models for flight delay prediction focus on distinguishing between delayed and on-time flights. Shau et al. [14] utilized a Support Vector Machine (SVM) to classify delays, effectively handling variables like weather, time of day, and seasonal patterns to achieve high classification accuracy. Rojas et al. [15] explored Random Forest classifiers for route-specific and seasonal delay prediction, showing improved accuracy by capturing route and seasonal variability. Ensemble-based methods such as Random Forest are noted for their robustness in high-dimensional data classification and have become a preferred choice in aviation-related studies due to their interpretability. In a novel combination, Wang et al. [16] integrated Naive Bayes classification with operational flight data to provide insights into delays caused by operational disruptions, achieving high accuracy in classifying delay status. Garcia et al. [17] employed a neural network-based classifier incorporating meteorological and operational variables to handle seasonal effects on delay patterns effectively. Li et al. [18] introduced the ST-RF model to improve departure delay prediction by integrating temporal and spatial dimensions. Their method systematically extracts geographic aspects of the aviation network using sophisticated network theory across multiple scales, including edge-, node-, and network-level characteristics. This comprehensive spatial analysis forms the foundation of their model's predictive capabilities.

2.2. Regression Approaches to Flight Delay Prediction

Regression models aim to estimate the exact duration of delays, offering quantitative insights that aid in real-time decision-making. Ye et al. [19] introduced a new approach that uses the LightGBM model to forecast aggregate flight departure delays. By leveraging airport-related aggregate characteristics, the methodology provides a comprehensive approach to enhancing the accuracy of flight departure delay forecasting. Mas-Pujol et al. [20] developed a CNN-LSTM hybrid for ATFM delay prediction, leveraging spatio-temporal features. Combining LSTM's capabilities to identify sequential relationships with CNN's abilities to extract spatial features, the model demonstrated high accuracy in delay duration prediction. Pineda et al. [21] applied a CNN-LSTM and Random Forest ensemble model to integrate weather, operational, and historical delay data, enhancing prediction accuracy

and interpretability. This approach highlighted the advantage of CNN’s spatial processing and LSTM’s time-series learning, leading to significant improvements in predictive power. Bisandu et al. [22] implemented an optimized CNN-LSTM model using a gradient-mayfly optimization algorithm, which proved particularly effective in high-traffic scenarios. The gradient-mayfly optimization allowed for fine-tuning parameters, improving delay prediction for flights impacted by dynamic traffic loads. To address temporal dependencies in delay data, Rojas et al. [23] employed an LSTM-based regression model with an attention mechanism. This model successfully captured delay propagation trends, offering accurate multi-step delay duration predictions. Random Forest has also been applied in regression contexts; Mamdouh et al. [24] used a multi-step Random Forest model that iteratively predicted delay durations by accounting for cumulative effects from prior flights. This model was beneficial for capturing cascading delays across interconnected flights, making it suitable for high-volume airports. Zhao et al. [25] applied a CNN with Random Forest regression for spatial–temporal delay prediction, achieving notable accuracy improvements by combining CNN’s spatial feature extraction with Random Forest’s robust regression capabilities. An overview of relevant studies on flight arrival delay, including its goals, modelling techniques, and factors taken into account, is presented in Table 1.

Table 1. An overview of recent studies on delay prediction.

Reference	Main Objective	Methodology	Considered Factors
Sahu et al. [14]	Binary delay prediction	SVM	Weather, time of day, seasonality
Rojas et al. [15]	Seasonal delay classification	Random Forest	Route, seasonal variability
Li et al. [18]	Flight delay prediction	ST-Random Forest	Temporal and spatial network features
Ye et al. [19]	Predicting the overall flight departure delay	Supervised learning (LightGBM)	Airport-related aggregate characteristics
Mas-Pujol et al. [20]	Delay duration prediction	CNN-LSTM	ATFM, spatio-temporal data
Pineda et al. [21]	Hybrid delay prediction	CNN-LSTM + RF	Weather, historical delays
Bisandu et al. [22]	High-traffic delay prediction	CNN-LSTM (optimized)	Gradient-mayfly tuning
Rojas et al. [23]	Temporal delay prediction	LSTM with attention	Sequential dependencies
Mamdouh et al. [24]	Multi-step delay prediction	Random Forest regression	Cumulative delay effects
Zhao et al. [25]	Spatial–temporal delay prediction	CNN + RF regression	Spatial, operational data

2.3. Our Proposed Model: Feature Engineering for Delay Prediction

Our proposed model emphasizes feature engineering, leveraging both temporal and spatial factors for improved delay prediction. We introduce a correlation-based feature selection (CFS) algorithm that ranks features by inter-feature correlations, retaining only those with high predictive potential and discarding redundancies. The DFFRN architecture is a five-layer deep neural network optimized for delay regression, comprising three hidden layers between input and output. This model enhances accuracy while reducing dimensionality, capturing the nonlinear dependencies present in delay data across spatio-temporal dimensions.

3. Methodology

Accuracy in flight delay prediction is vital for enhancing operational efficiency, improving passenger satisfaction, and reducing airline costs. It facilitates better air traffic management, informed travel decisions, timely updates, and effective scheduling. To achieve these goals, this research proposes comprehensive approaches aimed at significantly improving predictive performance by leveraging feature integration, feature engineering, and machine learning techniques, as shown in Figure 1. The process began with data preprocessing; then, we integrated essential geographical features, like the longitude and the latitude of an airport, into the dataset to enrich the predictive power of the models. We extracted extensive features, ensuring that all relevant temporal, spatial, and interaction features were captured. This was followed by Exploratory Data Analysis (EDA) to identify potential trends and patterns within the data. After the initial analysis, we employed novel feature

selection techniques, using correlation metrics to filter out redundant features. Notably, we developed a novel approach to select high-quality features that contributed most effectively to model performance. Following this, we used a variety of modelling techniques, such as ensemble-based models, tree-based models, deep learning models, and linear models.

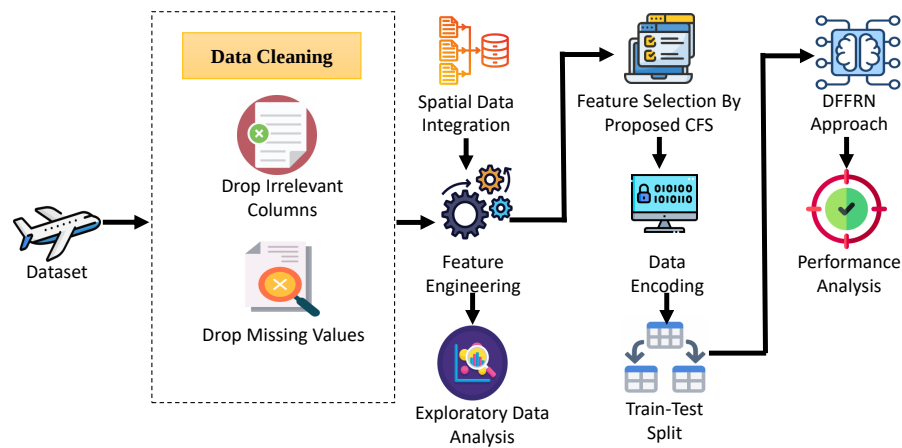


Figure 1. Overview of the methodology: This figure illustrates the step-by-step process that we followed, including data preprocessing, feature engineering, EDA, feature selection, and model building.

3.1. Data Cleaning

In the data cleaning phase, we handled missing values and dropped irrelevant columns.

3.1.1. Dropping Irrelevant Columns

As our objective is to predict arrival delays, the following columns are irrelevant because they capture information only available after the flight has landed or reached the destination, making them unsuitable for predicting pre-arrival delays. Table 2 summarizes these columns and the reasons for their exclusion.

Table 2. Summary of dropped columns and their irrelevance for predicting ARR_DELAY.

Column	Reason for Dropping
WHEELS_ON	Time when wheels touch the ground; provides post-arrival information irrelevant to predicting pre-arrival delays.
TAXI_IN	Time taken to taxi from landing to gate; useful for post-landing analysis but not for predicting arrival delays.
ELAPSED_TIME	Actual time taken from gate departure at origin airport to gate arrival at destination airport; provides post-flight data that does not contribute to predicting delays before arrival.
AIR_TIME	Total time in air; although it indicates flight progress, it occurs after departure, making it irrelevant for predicting pre-arrival delays.
ARR_TIME	Actual arrival time; directly tied to arrival delay, leading to potential data leakage if used in modeling.

3.1.2. Drop Cancellation and Diverted Data

To refine the dataset, we dropped flight data samples that were either canceled or diverted, as predicting delays requires focusing solely on operational flights where delay patterns can be observed. Mathematically, let D represent the original dataset containing N flights:

$$D = \{d_1, d_2, \dots, d_N\} \quad (1)$$

where each data point d_i represents a flight. We define two subsets:

- $C \subset D$: The subset of flights where each flight is canceled, i.e., $d_i \in C$ if CANCELLED is True.
- $V \subset D$: The subset of flights where each flight is diverted, i.e., $d_i \in V$ if DIVERTED is True.

The refined dataset D' , which excludes both canceled and diverted flights, is defined as follows:

$$D' = D \setminus (C \cup V) \quad (2)$$

This equation represents the new dataset D' , which contains all flights from D except those that are either canceled or diverted. After this filtering step, 3306 canceled flights and 549 diverted flights were removed.

3.1.3. Treating Missing Data

If missing information is not well treated, it can result in misleading estimates, decreased statistical accuracy, and incorrect conclusions. Based on the percentage of missing values in each column, different strategies were applied. Columns with a high percentage of missing values (more than 50%) were dropped from the dataset. Retaining these columns could introduce bias into the analysis, as the missing data might not be representative of the overall data distribution. According to Schafer and Graham (2002) [26], columns with more than 50% missing data are often removed to prevent potential biases and data distortions. Therefore, the following columns shown in Table 3 were removed from the dataset:

Table 3. Summary of dropped columns due to high missing values.

Column	Percentage of Missing Values (%)
CANCELLATION CODE	98.26
DELAY DUE CARRIER	76.81
DELAY DUE WEATHER	76.81
DELAY DUE NAS	76.81
DELAY DUE SECURITY	76.81
DELAY DUE LATE AIRCRAFT	76.81

For columns with a negligible percentage of missing values (defined as missing data points constituting up to 5% of the total entries, as suggested by Roderick J. A. Little [27]), rows containing missing values were removed to maintain data integrity. This approach is justified when the percentage of missing data is very low, as it minimizes the loss of valuable data while ensuring a complete dataset for analysis [27]. The following columns shown in Table 4 had negligible missing values and were handled by deleting the respective rows. After preprocessing, the dataset contained a total of 186,506 data points. To assess the missing data in rows, the pandas library was used. The info() function in pandas was employed to get an overview of the dataset. This allowed for a clear identification of columns with missing values. The rows with missing data were then removed using the dropna() function.

Table 4. Summary of dropped rows due to missing values.

Column	Percentage of Missing Values (%)
TAXI_OUT	1.74
WHEELS_OFF	1.74
DEP_DELAY	1.68
DEP_TIME	1.68
DEST_STATE	0.03
CRS_ELAPSED_TIME	0.00

After performing the data cleaning process, including removing irrelevant columns and rows with missing or inconsistent values, the dataset was refined. A sample of the

representative data after cleaning is shown in Table 5. These cleaned data now include only relevant and complete records for further analysis.

Table 5. Representative dataset after cleaning.

FL_DATE	AIRLINE	AIRLINE_CODE	DOT_CODE	ORIGIN	DEST	CRS_DEP_TIME	DEP_TIME	DEP_DELAY	TAXI_OUT	WHEELS_OFF	CRS_ARR_TIME	ARR_DELAY
9 January 2019	United Air Lines Inc.	UA	19977	FLL	EWR	1155	1151.0	−4.0	19.0	1210.0	1501	−14.0
19 November 2022	Delta Air Lines Inc.	DL	19790	MSP	SEA	2120	2114.0	−6.0	9.0	2123.0	2315	−5.0
22 July 2022	United Air Lines Inc.	UA	19977	DEN	MSP	954	1000.0	6.0	20.0	1020.0	1252	0.0
6 June 2023	Delta Air Lines Inc.	DL	19790	MSP	SFO	1609	1608.0	−1.0	27.0	1635.0	1829	24.0
23 February 2020	Spirit Air Lines	NK	20416	MCO	DFW	1840	1838.0	−2.0	15.0	1853.0	2041	−1.0

3.2. Incorporation of Airport Geographical Coordinates

To enhance our analysis of air delays, we integrated geographical coordinates (latitude and longitude) for both origin and destination airports by merging airport data (<https://pypi.org/project/airportsdata/> accessed on 6 December 2024, License: MIT License (The MIT License (MIT) Copyright (c) 2020-Mike Borsetti)). The transformation for adding geographical coordinates can be expressed mathematically. Let df_a represent the airport data and df_f the main dataset. The merging operation for origin and destination airports can then be defined as follows:

$$df_f = df_f \cup (df_a|_{\text{ORIGIN}}) \cup (df_a|_{\text{DEST}}) \quad (3)$$

In this equation, $df_f \cup df_a|_{\text{ORIGIN}}$ denotes merging the origin airport data with the flight dataset using the origin airport code as the key. Similarly, $df_f \cup df_a|_{\text{DEST}}$ denotes merging the destination airport data. This method ensures that each flight record contains the latitude and longitude for both the origin and destination airports, enabling spatial analysis of flight delays based on geographical factors.

3.3. Feature Engineering

A crucial step in predictive modeling is feature engineering, which entails developing new features or manipulating existing features to improve the model's capacity for prediction. By leveraging domain-specific expertise and the structure of the dataset, we developed several features aimed at capturing the temporal, spatial, and operational patterns in flight delays. This subsection elaborates on the temporal features derived from the dataset, providing insight into the rationale and methodology behind their creation.

3.3.1. Temporal Features

Temporal features are essential in understanding patterns related to time, such as flight schedules, peak hours, and operational timings, which significantly influence flight delays. We extracted several temporal features from the dataset, focusing on converting the scheduled departure time into useful components for analysis.

1. Month

The month of the flight was extracted to capture seasonal variations in flight delays. Delays can vary significantly between different months because of variables like weather conditions, holidays, and seasonal airport congestion. This feature is represented as an integer value ranging from 1 (January) to 8 (August).

The decision to focus on data from January to August was primarily driven by the scope of available and complete data at the time of analysis. This period covers diverse seasonal patterns, including winter, spring, and summer, offering a representative sample of different operational and weather-related conditions that impact flight delays. These months capture a range of significant events in aviation, such as peak winter travel, spring break, and summer vacation periods, which often exhibit distinct delay characteristics.

While excluding the remaining months (September to December) might appear as a limitation, it ensures the integrity of the analysis by using a fully validated and uniform dataset. Future work will aim to incorporate a full year of data to analyze trends comprehensively and compare seasonal variations across the entire calendar year.

2. Day of Month

This feature captures the specific day of the month when the flight occurs, allowing the model to identify potential patterns tied to certain dates, such as the beginning, middle, or end of the month when travel behaviors may vary.

3. Weekday

Weekday is extracted to capture patterns associated with specific weekdays. It is encoded as an integer from 0 (Monday) to 6 (Sunday). Flights on different days may exhibit different delay patterns; for instance, weekends and weekdays often have distinct traffic volumes and operational schedules.

4. Week of Year

This feature represents the ISO calendar week of the year for each flight. It helps capture seasonality and trends associated with specific weeks, such as holiday seasons or peak travel periods, which are likely to influence delay patterns.

5. Quarter

The quarter of the year (Q1, Q2, Q3, or Q4) is extracted to account for broader seasonal effects, such as differences between winter and summer or quarterly travel peaks. It ranges from 1 to 3 and provides a more generalized view of seasonality than the month feature.

6. Is Weekend

To differentiate between weekend and weekday flights, we created a binary feature that is set to 1 if the flight occurs on a Saturday or Sunday, and 0 otherwise. This feature captures differences in flight patterns, as weekends may experience different traffic volumes and operational schedules compared to weekdays. Mathematically, this feature is defined as follows:

$$I_w = \begin{cases} 1 & \text{if } w \geq 5 \\ 0 & \text{if } w < 5 \end{cases} \quad (4)$$

where I_w indicates whether the weekday w lies on a weekend (1) or not (0).

7. Hour Extraction

The time of day that the flight is scheduled was determined by extracting the hour component from CRS_DEP_TIME (formatted as HHMM). This makes it possible to examine delay patterns related to various times of the day, including the morning, afternoon, and evening.

8. Minute Extraction

We also extracted the minute component to provide more granular information. This helps capture variations within the same hour that may affect delays, such as congestion during specific periods of high traffic.

9. Departure Time Categorization

To analyze the effect of departure times on flight delays, we categorized the scheduled departure time (CRS_DEP_TIME) into predefined intervals. This variable, recorded in HHMM format, was divided into 12 time categories throughout the day. Each flight was assigned to one of these categories based on its scheduled departure time, simplifying the temporal analysis.

The categorization is mathematically represented as follows:

$$P = \begin{cases} 12 \text{ a.m.} - 2 \text{ a.m.} & \text{if } 0 \leq T < 200 \\ 2 \text{ a.m.} - 4 \text{ a.m.} & \text{if } 200 \leq T < 400 \\ 4 \text{ a.m.} - 6 \text{ a.m.} & \text{if } 400 \leq T < 600 \\ 6 \text{ a.m.} - 8 \text{ a.m.} & \text{if } 600 \leq T < 800 \\ 8 \text{ a.m.} - 10 \text{ a.m.} & \text{if } 800 \leq T < 1000 \\ 10 \text{ a.m.} - 12 \text{ p.m.} & \text{if } 1000 \leq T < 1200 \\ 12 \text{ p.m.} - 2 \text{ p.m.} & \text{if } 1200 \leq T < 1400 \\ 2 \text{ p.m.} - 4 \text{ p.m.} & \text{if } 1400 \leq T < 1600 \\ 4 \text{ p.m.} - 6 \text{ p.m.} & \text{if } 1600 \leq T < 1800 \\ 6 \text{ p.m.} - 8 \text{ p.m.} & \text{if } 1800 \leq T < 2000 \\ 8 \text{ p.m.} - 10 \text{ p.m.} & \text{if } 2000 \leq T < 2200 \\ 10 \text{ p.m.} - 12 \text{ a.m.} & \text{if } 2200 \leq T \leq 2359 \\ \text{Invalid Time} & \text{if } T > 2359 \end{cases} \quad (5)$$

where T is the scheduled departure time in HHMM format and P represents the corresponding time period of departure based on T .

10. Night Flight Indicator

Flights departing during the night or early morning hours may experience different patterns compared to daytime flights due to lower traffic levels or operational factors. To capture this effect, we created a binary feature called `IsNightFlight`, which is set to 1 if the scheduled departure time (`CRS_DEP_TIME`) is between 9:00 p.m. and 4:00 a.m., and 0 otherwise. This can be mathematically expressed as follows:

$$N = \begin{cases} 1 & \text{if } 2100 \leq T \leq 2359 \text{ or } 0 \leq T < 400 \\ 0 & \text{otherwise} \end{cases} \quad (6)$$

where T is the scheduled departure time in HHMM format, and N represents whether a flight is a night flight (1 if it departs between 9:00 p.m. and 4:00 a.m., 0 otherwise). The time range for night flights was set from 9:00 p.m. to 4:00 a.m. based on operational and behavioral patterns observed in aviation. This range aligns with common industry definitions of “red-eye” flights, which typically occur during late-night hours when passenger traffic and airport operations are lower. These hours often present distinct characteristics, such as reduced air traffic congestion, fewer operational staff, and varying weather conditions, which can influence delays differently compared to daytime flights. Additionally, the 9:00 p.m. to 4:00 a.m. window includes flights that depart in the late evening and arrive in the early morning, allowing the model to capture unique temporal trends associated with nighttime operations. By creating this binary feature, this study isolates potential patterns in delays specific to night operations, which might otherwise be obscured in a broader temporal analysis.

3.3.2. Previous Delay Extraction

1. Previous Departure Delay

The previous flight’s departure delay from the same origin airport is captured by this feature. It provides insight into recent performance and any recurring issues that may influence the current flight’s delay. Mathematically, it can be expressed as follows:

$$\Delta_{prev} = \Delta_{i-1} \quad \forall i \in \text{ORIGIN} \quad (7)$$

where the prior departure delay is represented by Δ_{prev} and the departure delay of flight i is represented by Δ .

2. Previous Arrival Delay

This feature records the arrival delay of the previous flight arriving at the same destination airport, helping to identify patterns related to arrival performance. It is represented as follows:

$$A_{prev} = A_{i-1} \quad \forall i \in \text{DEST} \quad (8)$$

where A is the arrival delay of flight i .

3. Rolling Departure Delay

The rolling departure delay feature computes the mean of the departure delays over the last 5 flights from the same origin airport. It provides a smoothed view of recent delay trends. The rolling average is defined as follows:

The average of the departure delays for the last five flights from the same origin airport is calculated using the rolling departure delay feature. It provides a smoothed view of recent delay trends. The rolling average is defined as follows:

$$\Delta_{roll} = \frac{1}{N} \sum_{k=i-N}^i \Delta_k \quad \forall i \in \text{ORIGIN} \quad (9)$$

where the number of prior flights taken into consideration is denoted by N , and Δ represents the departure delay of flight i . Δ_{roll} is the rolling average of the departure delays for the last N flights from the same origin.

4. Previous Rolling Departure Delay

This feature captures the average of the previous departure delays using a rolling window of the last 5 flights from the same origin airport. It smoothens the previous delay data for temporal analysis. It is expressed as follows:

$$\Delta_{prev_roll} = \frac{1}{M} \sum_{k=i-M}^{i-1} \Delta_{prev,k} \quad \forall i \in \text{ORIGIN} \quad (10)$$

where M is the number of prior flights taken into account, Δ_{prev} is the flight i 's previous departure delay, and Δ_{prev_roll} is the rolling average of the prior departure delays across the most recent M flights from the same origin.

5. Previous Rolling Arrival Delay

Similar to the rolling departure delay, this feature calculates the average of the previous arrival delays using a rolling window of the last 5 flights arriving at the same destination airport. It is defined as follows:

$$A_{prev_roll} = \frac{1}{N} \sum_{k=i-N}^{i-1} A_{prev,k} \quad \forall i \in \text{DEST}, N = 5 \quad (11)$$

The rolling average of the prior arrival delays to the same destination is represented by A_{prev} , the number of previous flights is represented by N , and the previous arrival delay of flight i is represented by A_{prev} .

6. Previous Arrival Delay Ratio

This feature computes the ratio of flights with previous arrival delays for each airline. It indicates how frequently an airline's flights arrive late based on historical data. The delay ratio is defined as follows:

$$R_{prev_arr} = \frac{\sum_{i=1}^m \mathbb{I}(A_{prev,i})}{m} \quad (12)$$

The overall quantity of flights is represented by m , the indicator function is represented by $\mathbb{I}$, the previous arrival delay is represented by A_{prev} , and the airline's ratio of prior arrival delays is represented by R_{prev_arr} .

3.3.3. Spatial Feature Extraction

Spatial features provide critical information about the geographical aspects of flights, such as the distance between the origin and destination airports. These features are essential for understanding how geographical factors impact flight operations and delays. One of the key spatial features extracted in this study is the haversine distance.

1. Haversine Distance

Using the latitude and longitude coordinates of two locations on the surface of the Earth, the haversine distance calculates the great-circle distance (d) between them. This feature captures the actual geographic distance between the origin and destination airports, which helps understand the correlation between distance and delays.

The haversine formula is defined as follows:

$$d = 2\mathcal{R} \cdot \arcsin \left(\sqrt{\sin^2 \left(\frac{\Delta\psi}{2} \right) + \cos(\psi_1) \cdot \cos(\psi_2) \cdot \sin^2 \left(\frac{\Delta\chi}{2} \right)} \right) \quad (13)$$

The great-circle distance between two Earth-surface sites is represented by the formula d . The Earth's radius is represented by the variable $\mathcal{R}$, the latitude difference is represented by $\Delta\psi = \psi_2 - \psi_1$, and the longitude difference is represented by $\Delta\chi = \chi_2 - \chi_1$.

2. Origin Cluster

We applied K-means clustering on the airports' latitudes and longitudes of origin to group them into 5 distinct clusters. This spatial grouping captures regional patterns in delays.

The clustering objective is mathematically represented as follows:

$$\text{minimize} \sum_{m=1}^n \sum_{j=1}^k \mathbb{I}(c_m = j) \cdot \|\mathbf{z}_m - \boldsymbol{\mu}_j\|^2 \quad (14)$$

The centroid of cluster j is represented by $\boldsymbol{\mu}_j$ and the feature vector for the m th airport by $\mathbf{z}_m$. The clusters that are produced are saved as `ORIGIN_cluster`.

3. Destination Cluster

Similarly, K-means clustering was applied to the latitude and longitude of destination airports, creating 5 clusters to capture spatial patterns influencing arrival delays. The mathematical representation is as follows:

$$\text{minimize} \sum_{i=1}^m \sum_{j=1}^k \mathbb{I}(c_i = j) \cdot \|\mathbf{y}_i - \boldsymbol{\mu}_j\|^2 \quad (15)$$

where the centroid of cluster j is represented by $\boldsymbol{\mu}_j$ and the feature vector for the i th destination airport is represented by $\mathbf{y}_i$. Clusters that are produced are saved as `DESTINATION_cluster`.

3.3.4. Interaction Features

Interaction features are created by combining two or more variables to capture complex relationships between them. These features help the model identify how different aspects of flight operations interact and potentially influence delays. The interaction features developed in our study are as follows:

1. Taxi-Out Time and Departure Delay Interaction

The product of taxi-out time and departure delay is used to compute this characteristic, which is represented as I_{TDD} (Interaction of Taxi-Out Time and Departure Delay). It captures the relationship between ground movement time and departure performance:

$$I_{TDD} = T_{\text{out}} \times \Delta_{\text{dep}} \quad (16)$$

where Δ_{dep} is the delay before takeoff and T_{out} is the time the aircraft takes to taxi to the runway.

2. Interaction Between Distance and Departure Delay

This feature, I_{DD} (Interaction of Distance and Departure Delay), represents the interaction between flight distance and departure delay. It helps analyze whether the length of a flight impacts the extent of the departure delay:

$$I_{DD} = D \times D_{\text{dep}} \quad (17)$$

where d is the geographic distance between the airports of origin and destination, and Δ_{dep} represents the departure delay.

3. Interaction Between Distance and Arrival Delay

This feature, I_{DAD} (Interaction of Distance and Arrival Delay), is the product of flight distance d and arrival delay A_{arr} . It examines whether longer flight distances correlate with greater or reduced arrival delays:

$$I_{DAD} = d \times A_{\text{arr}} \quad (18)$$

where A_{arr} is the delay incurred upon arrival at the destination airport.

3.3.5. Additional Features

1. Is Departure Delayed

The model's capacity to identify patterns in the data is further improved using additional features that offer binary or categorical information. I_{dep} (Is Departure Delayed) is a binary characteristic that indicates if a flight had a delay in departure. If departure delay Δ_{dep} is larger than zero, it is set to 1; otherwise, it is set to 0:

$$I_{\text{dep}} = \begin{cases} 1 & \text{if } \Delta_{\text{dep}} > 0 \\ 0 & \text{otherwise} \end{cases} \quad (19)$$

where I_{dep} is an indicator variable that equals 1 if there is a departure delay ($\Delta_{\text{dep}} > 0$), and 0 otherwise.

2. Cyclical Encoding of Departure Hour

Time, especially hours of the day, is cyclical (e.g., 11 p.m. is closer to midnight than it is to noon). To capture this cyclical nature in our model, we transformed the departure hour (DEP_HOUR) into sine and cosine values. This method preserves the cyclical relationship, ensuring that adjacent hours are represented similarly in the model. The transformation is defined as follows:

$$S = \sin\left(\frac{2\pi \times H}{24}\right) \quad (20)$$

$$C = \cos\left(\frac{2\pi \times H}{24}\right) \quad (21)$$

where H is the departure hour, S represents the sine-transformed value of the departure hour, and C represents the cosine-transformed value of the departure hour.

3. Departure Delay Ratio

This feature measures the proportion of flights that have experienced departure delays for each airline, based on current departure delay values. It is mathematically expressed as follows:

$$R_{\text{dep}} = \frac{\sum_{i=1}^m \mathbb{I}(\Delta_i)}{m} \quad (22)$$

The entire quantity of flights is denoted by m , and the departure delay of flight i is represented by Δ_i .

4. Previous Departure Delay Ratio

This feature calculates the ratio of flights with previous departure delays for each airline. It provides insights into the historical performance of airlines regarding departure punctuality. The formula is the following:

$$R_{prev_dep} = \frac{\sum_{i=1}^m \mathbb{I}(\Delta_{prev,i})}{m} \quad (23)$$

where the number of flights is m and the prior departure delay of flight i is represented by $\Delta_{prev,i}$.

3.4. Exploratory Data Analysis

In the phase of exploratory data analysis (EDA), we analyzed the dataset to uncover key patterns in flight delays. This included examining flights per state, the relationship between departure and arrival delays, seasonal trends, weekday vs. weekend flight patterns, hourly trends, and day vs. night trends.

3.4.1. Flights per State

We analyze the frequency of flights originating from different states in the United States in 2023, focusing on California, Texas, Florida, New York, and Georgia. To analyze the distribution of flights per state, we group the data by ORIGIN_STATE to determine how many flights there were in each state overall:

$$F(s) = \sum_{i=1}^n \delta(O_i = s) \quad (24)$$

where $F(s)$ is the flight count for state s , O_i represents the origin state of flight i , and δ is the indicator function:

$$\delta(O_i = s) = \begin{cases} 1, & \text{if } O_i = s \\ 0, & \text{otherwise} \end{cases} \quad (25)$$

The bar chart shows these counts $F(s)$ per state, highlighting absolute differences in flight volume. The pie chart depicts the relative proportion of flights from each state, calculated as follows:

$$P(s) = \frac{F(s)}{\sum_s F(s)} \times 100 \quad (26)$$

where $P(s)$ represents the percentage of total flights from state s . This dual visualization provides both absolute and proportional insights into state-level flight distribution.

3.4.2. Departure and Arrival Delay Relationship

The relationship between average departure delay $\bar{\Delta}$ and average arrival delay $\bar{Arr}$ for each airline is analyzed using a scatter plot. The average delays are computed as follows:

$$\bar{Arr} = \frac{1}{M} \sum_{i=1}^M Arr_i \quad (27)$$

$$\bar{\psi} = \frac{1}{Z} \sum_{i=1}^Z \psi_i \quad (28)$$

where M , Z is the total number of flights for each airline, and Arr_i and ψ_i stand for the arrival and departure delays of flight i , respectively.

To examine the correlation between the average delays, a trend line was fitted to the scatter plot data, represented by the following equation:

$$y = px + q \quad (29)$$

where (p) is the line's slope, (q) is the y-intercept, (y) is the average arrival delay, and (x) is the average departure delay. This correlation aids in determining whether lengthier arrival delays are typical for airlines with higher departure delays.

3.4.3. Seasonal Trend

The analysis of flight counts per season is performed by aggregating the data based on the season of each flight. The total number of flights for each season s is calculated using the following equation:

$$F(s) = \sum_{i=1}^N \delta(S_i = s) \quad (30)$$

where $F(s)$ represents the total number of flights in season s , S_i denotes the season of flight i , and δ is the indicator function, which is defined as follows:

$$\delta(S_i = s) = \begin{cases} 1, & \text{if } S_i = s \\ 0, & \text{otherwise} \end{cases} \quad (31)$$

3.4.4. Weekend vs. Weekday Flight Analysis

The analysis focuses on the relationship between flight counts and average arrival delays categorized by weekdays and weekends. The total flight count for each category is computed using the following equation:

$$F(w) = \sum_{i=1}^M \alpha(W_i = w) \quad (32)$$

where the total number of flights on a given day is shown by $F(w)$, where w is either "Weekday" or "Weekend", W_i represents the day type of flight i , and α is the indicator function defined as follows:

$$\delta(W_i = w) = \begin{cases} 1, & \text{if } W_i = w \\ 0, & \text{otherwise} \end{cases} \quad (33)$$

The average arrival delay for each category is calculated using the following:

$$A(w) = \frac{1}{F(w)} \sum_{i=1}^N A_i \cdot \delta(W_i = w) \quad (34)$$

where $A(w)$ denotes the average arrival delay for the category w , and A_i is the arrival delay of flight i .

3.4.5. Flight Analysis by Hour

The analysis of average flight counts based on departure hours and days of the week involves aggregating flight data to identify trends and patterns. The total number of flights for each combination of departure hour and day of the week is computed as follows:

$$N(h, d) = \sum_{i=1}^M \delta(H_i = h, W_i = w) \quad (35)$$

where $N(h, d)$ represents the total flight count for hour h and day d , H_i is the departure hour of flight i , W_i denotes the day of the week for flight i , and δ is the indicator function defined as follows:

$$\delta(H_i = h, W_i = w) = \begin{cases} 1, & \text{if } H_i = h \text{ and } W_i = w \\ 0, & \text{otherwise} \end{cases} \quad (36)$$

After calculating the flight counts, a line plot is generated to visualize the average flight counts across all hours of the day. The average flight count for each departure hour is represented as follows:

$$A(h) = \frac{N(h)}{C(h)} \quad (37)$$

where $A(h)$ is the average flight count for hour h , $N(h)$ is the total number of flights during that hour, and $C(h)$ is the count of unique days observed during that hour.

3.4.6. Day vs. Night Flight Analysis

The analysis begins by aggregating flight data based on whether flights occur during night hours, represented by the binary variable I . The first step involves calculating the total flight count for both night and day flights:

$$N_0 = \sum_{i=1}^M \delta(I_i = 0) \quad (38)$$

$$N_1 = \sum_{i=1}^M \delta(I_i = 1) \quad (39)$$

where N_0 and N_1 represent the flight counts for non-night and night flights, respectively, and the overall quantity of flights is M .

Next, the average arrival delays for each category are calculated:

$$\text{avg_delay}_0 = \frac{1}{N_0} \sum_{i=1}^M \text{ARR_DELAY}_i \cdot \delta(I_i = 0) \quad (40)$$

$$\text{avg_delay}_1 = \frac{1}{N_1} \sum_{i=1}^M \text{ARR_DELAY}_i \cdot \delta(I_i = 1) \quad (41)$$

where avg_delay_0 and avg_delay_1 are the average arrival delays for non-night and night flights, respectively.

To analyze the overall distribution of flights, the percentages of night and non-night flights are computed:

$$P_0 = \frac{N_0}{N_0 + N_1} \times 100 \quad (42)$$

$$P_1 = \frac{N_1}{N_0 + N_1} \times 100 \quad (43)$$

where P_0 and P_1 are the percentages of day and night flights, respectively.

3.5. Data Encoding

To prepare the dataset for model training, we applied different encoding techniques to handle categorical and numerical values effectively. For categorical features, we used one-hot encoding for deep learning models to allow them to process binary representations of each category, enhancing model interpretability. In contrast, for traditional machine learning models, we used label encoding, which gives each category a distinct number label, improving the data's suitability for algorithms that do not need binary representations.

To guarantee that every numerical value had a mean of 0 and a standard deviation of 1, we used standard scaling for numerical characteristics. By keeping features with bigger sizes as a result of controlling the training process, this normalization phase speeds up convergence and increases stability for models.

3.6. Train-Test Split

The dataset, consisting of a total of 186,506 data points, was separated into sets for testing and training in order to make model creation and assessment easier. The data

were split 80/20, with 20% going to testing and 80% going to training. As a consequence, the training set contained 149,205 data points, while the testing set contained 37,301 data points. Such a partitioning ensures a robust dataset size for training, while retaining sufficient data for an unbiased evaluation of the model's performance.

3.7. Proposed Feature Selection Algorithm

We proposed a feature selection process for our system. The detailed algorithm presented in Algorithm 1 begins by computing the absolute correlation matrix for all features in the dataset, denoted as $\mathbf{C}$. From this matrix, each feature's correlation values with the target feature (T) are extracted and features that meet the specified correlation threshold τ are filtered. These features are stored in the set F_{filtered} , excluding the target feature itself. Features are chosen by the feature selection algorithm according to how well they correlate with the desired attribute. Prior to filtering characteristics with correlations higher than a predetermined threshold, it computes the correlation matrix. Then, in order to reduce redundancy and preserve the most relevant features, it eliminates features that have almost exact correlations to the target.

Algorithm 1 Feature Selection Based on Correlation

Input: Dataset D , Target feature T , Correlation threshold τ

Output: Selected subset of features F_{selected}

- 1: Compute the absolute correlation matrix $\mathbf{C}$ for all features in D
- 2: Determine the correlation coefficients between the target feature T and each feature f_i :

$$C_T = \{\text{cor}(f_i, T) \mid f_i \in D\}$$

- 3: Filter features based on the correlation threshold τ :

$$F_{\text{filtered}} = \{f_i \mid C_T(f_i) \geq \tau \text{ and } f_i \neq T\}$$

- 4: Initialize an empty set for selected features: $F_{\text{selected}} \leftarrow \emptyset$
 - 5: **for** each feature f_i in F_{filtered} **do**
 - 6: Set $\text{keep_feature} \leftarrow \text{True}$
 - 7: **for** each feature f_j where $j > i$ in F_{filtered} **do**
 - 8: **if** $|C_T(f_i) - C_T(f_j)| \leq \epsilon$ (e.g., 0.0005) **then**
 - 9: **if** $C_T(f_i) < C_T(f_j)$ **then**
 - 10: Set $\text{keep_feature} \leftarrow \text{False}$
 - 11: **break**
 - 12: **end if**
 - 13: **end if**
 - 14: **end for**
 - 15: **if** keep_feature is **True** **then**
 - 16: Add f_i to F_{selected}
 - 17: **end if**
 - 18: **end for**
 - 19: **Return** F_{selected}
-

To ensure the selected features are not redundant, the algorithm compares the correlation values between pairs of features in F_{filtered} . For each feature f_i , it checks whether the correlation difference with another feature f_j is below a small threshold ϵ , and only retains the feature with the higher correlation to the target. This process ensures that only the most relevant features with minimal redundancy are added to the final set of selected features, F_{selected} .

3.8. The Architecture of the Suggested Model

The suggested modeling approach's architecture, named the Deep Feed Forward Regression Network (DFFRN) for regression, is constructed as a series of fully connected layers. The final prediction is reached when the input is successively changed using linear operations and nonlinear activation functions. It then moves through three hidden layers of 128, 64, and 32 neurons, respectively. The following Figure 2 illustrates the architecture of the proposed DFFRN model.

The algorithm for training our proposed DFFRN is represented in Algorithm 2, which initializes the input parameters, including the training data X_{train} and target y_{train} . The following layers are added to a sequential model $\mathcal{M}$: an input layer, followed by dense layers with 128, 64, and 32 units using ReLU activation. The Adam optimizer compiles the model and defines the Mean Squared Error (MSE) loss function.

Algorithm 2 Neural Network Training Algorithm

- 1: **Input:** Training data $X_{\text{train}} \in \mathbb{R}^{n \times m}$, target $y_{\text{train}} \in \mathbb{R}^n$, number of epochs E , batch size B
- 2: Initialize modeling approach $\mathcal{M}$
- 3: Add input layer's shape (m)
- 4: Add a dense layer with ReLU activation and 128 units:

$$Z^{(1)} = W^{(1)} \cdot X_{\text{train}} + b^{(1)} \quad \text{and} \quad A^{(1)} = f(Z^{(1)})$$

- 5: Apply a 64-unit dense layer with ReLU activation:

$$Z^{(2)} = W^{(2)} \cdot A^{(1)} + b^{(2)} \quad \text{and} \quad A^{(2)} = f(Z^{(2)})$$

- 6: Add dense layer with 32 units and ReLU activation:

$$Z^{(3)} = W^{(3)} \cdot A^{(2)} + b^{(3)} \quad \text{and} \quad A^{(3)} = f(Z^{(3)})$$

- 7: Add output layer with 1 unit:

$$Y_{\text{pred}} = W^{(4)} \cdot A^{(3)} + b^{(4)}$$

- 8: Define the loss function:

$$\mathcal{L}(Y_{\text{pred}}, y_{\text{train}}) = \frac{1}{m} \sum_{j=1}^m (y_j - Y_{\text{pred},j})^2$$

- 9: Compile model $\mathcal{M}$ with optimizer:

$$\mathcal{M}.\text{compile}(\text{optimizer} = \text{Adam}, \text{loss} = \mathcal{L})$$

- 10: **Initialize:** Early stopping criteria with patience $P \leftarrow 10$

- 11: **Train the model:**

- 12: **for** $epoch \leftarrow 1$ to E **do**

- 13: **for** each batch B_i in X_{train} **do**

- 14: Compute gradients $\nabla \mathcal{L}$ using backpropagation

- 15: Adjust the weights:

$$Weight^{(l)} \leftarrow Weight^{(l)} - \eta \nabla \mathcal{L}(Weight^{(l)})$$

- 16: Adjust biases:

$$bias^{(l)} \leftarrow bias^{(l)} - \eta \nabla \mathcal{L}(bias^{(l)})$$

- 17: **end for**

- 18: Monitor validation loss

- 19: **if** the validation loss for P epochs does not decrease **then**

- 20: **break** training

- 21: **end if**

- 22: **end for**

- 23: **Output:** Trained model $\mathcal{M}$
-

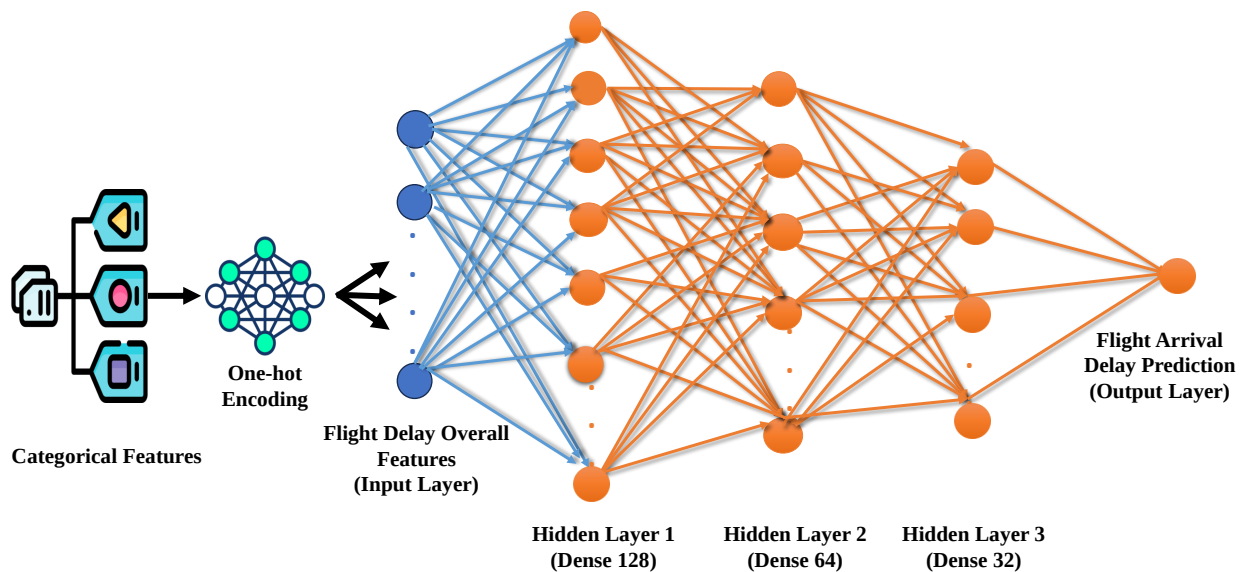


Figure 2. Proposed Deep Feedforward Regression Network (DFFRN) model structure with specified hidden layers.

The Deep Feed Forward Regression Network (DFFRN) architecture proposed in this study demonstrates significant advantages over traditional models, including superior predictive accuracy, robust handling of spatio-temporal features, and enhanced ability to identify complex nonlinear relationships in flight delay data. These strengths position DFFRN as a powerful tool for delay prediction and resource optimization in the aviation industry. However, like any advanced modeling approach, it is important to acknowledge potential limitations while highlighting how its design and implementation mitigate these challenges.

The primary strength of DFFRN lies in its ability to handle large-scale, high-dimensional data efficiently through its multi-layer architecture. The inclusion of carefully engineered spatio-temporal features and the correlation-based feature selection (CFS) algorithm ensures that only the most relevant and non-redundant features are utilized, minimizing noise and enhancing computational efficiency. Additionally, the deep architecture is capable of capturing both linear and nonlinear dependencies, which are often present in flight delay datasets, providing predictive accuracy that significantly outperforms traditional methods such as Random Forest, LSTM, or XGBoost.

Furthermore, the model incorporates advanced training techniques, such as early stopping and adaptive learning rates, to prevent overfitting. These techniques ensure that the model generalizes well to unseen data, even when applied to smaller datasets. The DFFRN's scalability also makes it adaptable to varying data sizes and complexities, allowing it to be tailored for broader applications in different domains. The potential limitations of the proposed DFFRN are discussed below:

a. Computational Cost:

Deep learning models, including DFFRN, inherently involve higher computational demands due to their multi-layer architecture and extensive training requirements. However, the modular nature of DFFRN allows it to be parallelized and optimized for high-performance computing environments, reducing training time significantly. Leveraging cloud-based platforms or hardware accelerators such as GPUs can further mitigate computational overhead.

b. Sensitive Hyperparameters:

Like most machine learning models, DFFRN relies on hyperparameter tuning for optimal performance. To address this, automated methods such as grid search or Bayesian optimization can be employed to systematically identify the best parameter configurations.

c. Risk of Overfitting in Smaller Datasets:

While the model is highly effective with large datasets, overfitting may occur when applied to smaller datasets. To counter this, regularization techniques such as dropout and L2 regularization have been incorporated into the architecture to reduce model complexity and enhance generalization.

d. Exploration of Ensemble Techniques:

To further enhance performance and mitigate potential limitations, ensemble methods such as combining DFFRN with tree-based models like Random Forest or gradient boosting could be explored. These methods can provide complementary strengths, leading to even higher accuracy and robustness.

While potential limitations of DFFRN exist, its advantages in accuracy, scalability, and handling of complex relationships far outweigh these challenges. By leveraging advanced training techniques, scalable computing solutions, and ensemble methods, the DFFRN architecture can be optimized for diverse applications, ensuring its utility and reliability across various operational contexts.

3.9. Baseline Models

For comparison, we use a range of baseline models, such as those derived from deep learning, ensemble techniques, tree-based methods, and linear regression.

1. Linear Regression

The relationship between the independent variables X and the dependent variable y is modeled using linear regression [28] as follows:

$$y = \alpha_0 + \alpha_1 X_1 + \alpha_2 X_2 + \dots + \alpha_m X_m + \epsilon \quad (44)$$

where α_i are the coefficients representing the impact of each feature, and ϵ is the error term.

2. Decision Tree

DT [29] recursively creates subgroups from the feature space based on feature values, represented as follows:

$$f(X) = \begin{cases} c_1, & \text{if } X \text{ is in region } R_1, \\ c_2, & \text{if } X \text{ is in region } R_2, \\ \vdots & \\ c_k, & \text{if } X \text{ is in region } R_k. \end{cases} \quad (45)$$

where c_i are the class labels or values assigned to regions R_i .

3. Random Forest (RF)

The ensemble technique called RF [30] builds many decision trees and combines their predictions:

$$f(X) = \frac{1}{N} \sum_{j=1}^N f_j(X) \quad (46)$$

where $f_j(X)$ is the prediction from the j -th tree, and N is the number of trees.

4. Bagging Regressor (BR)

BR [31] combines multiple base learners to improve stability and accuracy:

$$f(P) = \frac{1}{K} \sum_{i=1}^K f_i(P) \quad (47)$$

where $f_i(P)$ are the predictions of the basis learners and K is the number of base learners.

5. CatBoost

CatBoost [32] uses categorical features directly and symmetrically constructs decision trees:

$$f(P) = \sum_{m=1}^M \delta_m \phi_m(P) \quad (48)$$

where $\phi(P)$ are the base trees, and γ_m are their respective weights.

6. XGBoost

XGBoost [33] optimizes gradient boosting using a regularization term:

$$L = \sum_{j=1}^m \ell(z_j, f(x_j)) + \sum_{m=1}^M \phi(\psi_m) \quad (49)$$

where ϕ is a regularization term that penalizes complexity, and ℓ is the loss function.

7. Histogram-Based Gradient Boosting Regressor (HGBR)

HGBR [34] builds trees efficiently using histograms of feature values:

$$f(P) = \sum_{n=1}^N \delta_n \psi_n(P) \quad (50)$$

where δ_n are the weights of the weak learners in the ensemble and $\psi_n(X)$ are the weak learners.

8. AdaBoost

AdaBoost [35] combines weak learners, assigning weights based on their performance:

$$f(P) = \sum_{i=1}^I \phi_i \psi_i(P) \quad (51)$$

where each weak learner's weight is represented by $\phi_i \psi_i(P)$.

9. Gradient Boosting Regressor

In a sequential model-building process, GBR [36] fixes the errors of its predecessors:

$$y(P) = y_0(P) + \sum_{l=1}^L \delta_l \psi_l(P) \quad (52)$$

where the starting prediction is represented by $y_0(P)$, and the learning rate and weak learners are represented by δ_l and $\psi_l(P)$, respectively.

10. LightGBM Regressor (LGBMR)

LightGBM [37] uses a histogram-based approach to speed up training:

$$z(P) = \sum_{c=1}^C \alpha_c \delta_c(P) \quad (53)$$

where the base learners are denoted by $\delta_c(P)$ and their corresponding weights by α_m .

11. Multilayer Perceptron (MLP)

An MLP [38] computes its output as follows:

$$M^{(l)} = f(\text{weight}^{(l)} M^{(l-1)} + \text{bias}^{(l)}) \quad (54)$$

where the activation of the l -th layer is represented by $M^{(l)}$, with weights $\text{Weight}^{(l)}$ and biases $\text{bias}^{(l)}$.

12. Convolutional Neural Network (CNN)

CNNs [39] process grid-like data (e.g., images) using convolutional layers:

$$F_{\text{pred}} = f(P; \text{Weight}, \text{bias}) = C(P, \text{Weight}) + \text{bias} \quad (55)$$

where the convolution process with weights and biases is represented by C .

13. Long Short-Term Memory (LSTM)

LSTMs [40] are specialized RNNs designed for sequence prediction, with the cell state M_t and hidden state k_t updated as follows:

$$p_t = \zeta(Q_p \cdot [r_{t-1}, S_t] + c_p) \quad (56)$$

$$j_t = \zeta(Q_j \cdot [r_{t-1}, S_t] + c_j) \quad (57)$$

$$M_t = j_t \cdot M_{t-1} + p_t \cdot \tilde{M}_t \quad (58)$$

$$k_t = w_t \cdot \tanh(M_t) \quad (59)$$

where p_t, j_t, w_t are the input, forget, and output gates, Q and c are weights and biases, and ζ is a sigmoid-like function.

4. Result and Discussion

4.1. Data Description

The dataset for this study was derived from publicly available flight data provided by the Bureau of Transportation Statistics (BTS) for the year 2023 (<https://www.kaggle.com/datasets/patrickzel/flight-delay-and-cancellation-dataset-2019-2023/data>, accessed on 7 December 2024). The data include detailed flight records for five major U.S. states: California, Texas, Florida, New York, and Georgia. BTS is a reliable and comprehensive source of aviation data, ensuring the credibility of the information used. Additionally, geographical data, such as latitude and longitude for origin and destination airports, were integrated from the Airports Data Package, which is licensed under the MIT license and provides accurate spatial information for analytical purposes. The dataset includes a combination of categorical, numerical, temporal, and spatial attributes that describe flight operations. Key attributes in the dataset are given below:

- Flight Details:
 - Flight date, flight number, airline name, and Department of Transportation (DOT) codes.
 - Origin and destination airport codes, along with city and state names.
- Operational Data:
 - Scheduled and actual departure and arrival times (in HHMM format).
 - Delays in minutes for departure and arrival (e.g., DEP_DELAY, ARR_DELAY).
 - Taxi times (TAXI_OUT, TAXI_IN), and air time (AIR_TIME).
- Categorical Information:

- Indicators for cancellations (CANCELLED), diversions (DIVERTED), and reasons for delays (e.g., due to weather, late aircraft).
- Airport-level information, such as geographical coordinates and clustering based on origin and destination.
- Spatial–Temporal Features:
 - Derived features, such as haversine distance between origin and destination, clustering information, departure time categories, and weekend indicators.
 - Temporal patterns (e.g., month, day of week, hour of day) to capture seasonality and hourly trends in delays.

Data cleaning was performed to ensure the quality and relevance of the dataset. This involved the following:

- Removing flights that were canceled or diverted to focus on operational flights with meaningful delay patterns.
- Dropping columns with excessive missing values or those irrelevant to pre-arrival delay predictions (e.g., post-arrival attributes such as ARR_TIME).
- Imputing or removing rows with negligible missing data, ensuring data integrity without introducing bias.

To enrich the dataset, we incorporated spatial coordinates for origin and destination airports. This integration enabled the calculation of haversine distance and clustering of airports into regions to capture spatial dependencies. Additionally, derived features, such as rolling averages for departure and arrival delays, were calculated to enhance the model’s predictive capability.

The dataset contains various features related to flight information, delays, and airline operations. Table 6 provides a summary of each feature, detailing its role in capturing aspects of flight scheduling, delay sources, and operational specifics.

Table 6. Description of dataset features.

Feature	Description
FL DATE	The flight date
AIRLINE	The name of the airline
AIRLINE DOT	The airline’s Department of Transportation (DOT) code
AIRLINE CODE	The airline’s specific code
DOT CODE	Code given by the Department of Transportation
FL NUMBER	Distinct flight number
ORIGIN	The code for the origin airport
ORIGIN_CITY	The name of the origin city
DEST	The code for the destination airport
DEST_CITY	The name of the destination city
CRS DEP TIME	The time of scheduled departure
DEP TIME	The actual time of departure
DEP DELAY	Minutes of departure delay
TAXI OUT	The duration of time it took to taxi out of the gate
WHEELS OFF	The moment the aircraft lifted off the ground
WHEELS ON	The moment the aircraft landed
TAXI IN	The duration of time needed to taxi into the gate
CRS ARR TIME	Arrival time assigned
ARR TIME	The time of arrival in reality
ARR DELAY	Minutes of arrival delay
CANCELLED	Shows if the flight was cancelled (1 for cancelled flights, 0 for other flights).
CANCELLATION_CODE	Code of reason for cancellation
DIVERTED	Shows whether the flight was rerouted (1 if it was, 0 otherwise).
CRS ELAPSED TIME	The flight’s scheduled elapsed time
ELAPSED TIME	Actual time that has passed

Table 6. Cont.

Feature	Description
AIR TIME	Total time spent in air
DISTANCE	Distance between airports of origin and destination
DELAY DUE CARRIER	Delay brought on by problems unique to airline
DELAY DUE WEATHER	Weather-related delays
DELAY DUE NAS	National Air System (NAS) caused delay.
DELAY DUE SECURITY	Delay caused due to problems with security
DELAY DUE LATE AIRCRAFT	Delay caused by aircraft's delayed arrival

4.2. Performance Metrics

Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and the coefficient of determination (R^2) were the three main metrics we used to assess the effectiveness of the flight delay prediction models. Every indicator sheds light on the robustness and forecasting accuracy of the models.

4.2.1. Root Mean Square Error (RMSE)

Higher errors are penalised by the RMSE [41], which calculates the average magnitude of error between anticipated and actual values. It is described as follows:

$$\text{Error}_{\mathcal{L}^2} = \sqrt{\frac{1}{\kappa} \sum_{v=1}^{\kappa} (\delta_v - \hat{\delta}_v)^2} \quad (60)$$

where κ is the total number of data points, δ_v is the actual value, and $\hat{\delta}_v$ is the forecast value. Better model accuracy is indicated by a smaller $\text{Error}_{\mathcal{L}^2}$.

4.2.2. Mean Absolute Error (MAE)

The mean absolute variance between expected and actual values is provided by the MAE [42], which offers a clear interpretation of model correctness. It is computed as follows:

$$\mathcal{A} = \frac{1}{v} \sum_{\theta=1}^v |\gamma_{\theta} - \hat{\gamma}_{\theta}| \quad (61)$$

where the actual value is denoted by γ_{θ} , the projected value by $\hat{\gamma}_{\theta}$, and the total number of data points by v . Better model accuracy is indicated by a smaller $\mathcal{A}$.

4.2.3. Coefficient of Determination R^2

The coefficient of determination, or R^2 metric [43], measures the percentage of variation in the actual values that can be predicted from the model. It is provided by the following:

$$\mathcal{R}^2 = 1 - \frac{\sum_{\zeta=1}^{\xi} (\omega_{\zeta} - \hat{\omega}_{\zeta})^2}{\sum_{\zeta=1}^{\xi} (\omega_{\zeta} - \bar{\omega})^2} \quad (62)$$

where the mean of the actual values is represented by $\bar{\omega}$. A model with high prediction accuracy is indicated by a $\mathcal{R}^2$ value closer to 1 as it shows that a significant amount of the variance is captured by the model.

4.3. Exploratory Data Analysis Results

4.3.1. Flights per State

In 2023, a total of 186,506 flights originated from the five aforementioned major states, as illustrated in Figure 3. California (CA) leads with 49,255 flights, surpassing Texas (TX) by 4%, and Florida (FL) by 21%. New York (NY) trails behind Florida with 38% fewer flights, while Georgia (GA) recorded 4% fewer flights than New York.

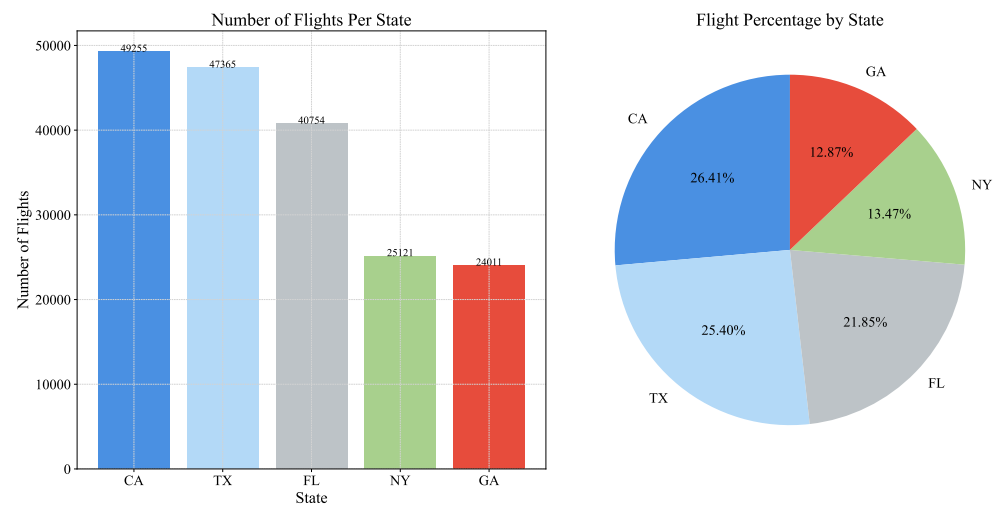


Figure 3. Frequency of flights per state in 2023.

4.3.2. Departure and Arrival Delays Analysis

In the analysis of average delays for various airlines, we observe distinct patterns in arrival and departure delays for the year 2023, as illustrated in Figure 4. Frontier Airlines (F9) exhibits the highest average arrival delay at 24.70 min, closely followed by Spirit Airlines (NK) with 21.58 min. Both airlines also show considerable departure delays, indicating potential operational challenges. In contrast, Alaska Airlines (AS) demonstrates the least average arrival delay at 3.25 min, coupled with a departure delay of 7.18 min, suggesting effective operational performance. Additionally, a linear relationship can be observed between average departure and arrival delays across the airlines; as departure delays increase, arrival delays tend to rise as well, indicating that timely departures are crucial for minimizing overall delays.

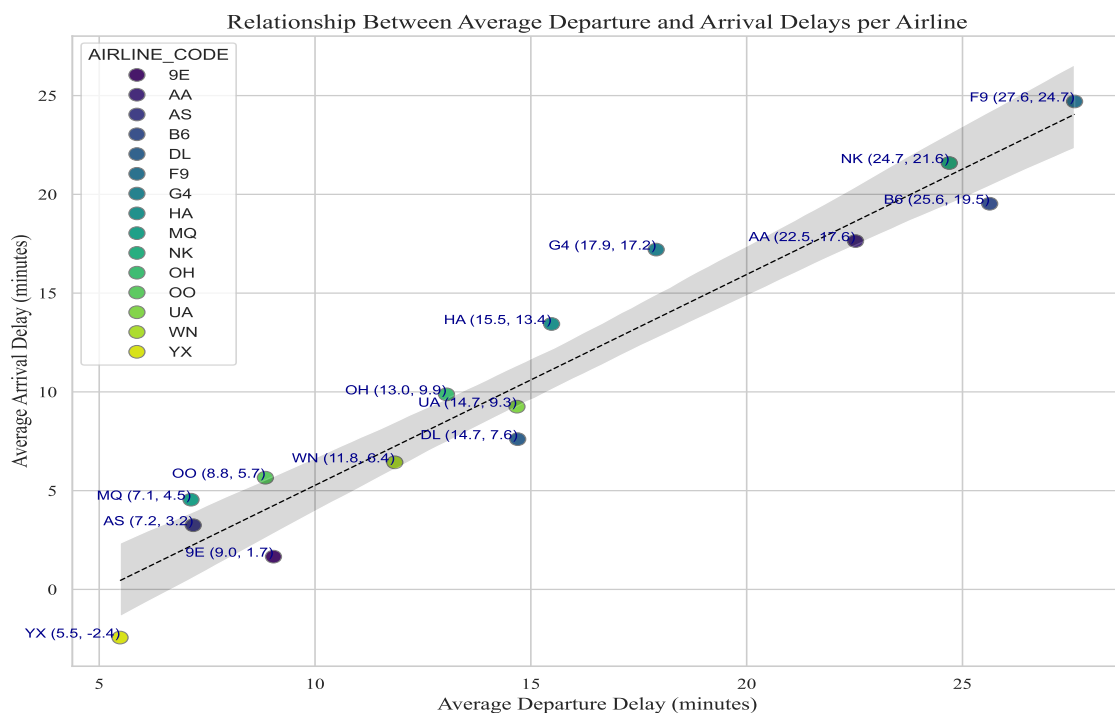


Figure 4. Arrival and departure delay relationship for airlines.

4.3.3. Seasonal Trend Analysis

In the analysis of flight counts, as shown in Figure 5, for the year 2023, significant seasonal variations are observed. The Spring season recorded the highest number of flights, totaling 70,788 flights, indicating a strong demand for air travel during this period. In contrast, the Summer season experienced a notable decrease, with 48,507 flights, reflecting typical patterns of reduced air traffic as families often take vacations. The Winter season accounted for 66,864 flights showcasing moderate travel activity, likely influenced by holiday travel and winter sports. These results highlight the fluctuations in air traffic throughout the year, with Spring being the most active season for flights.

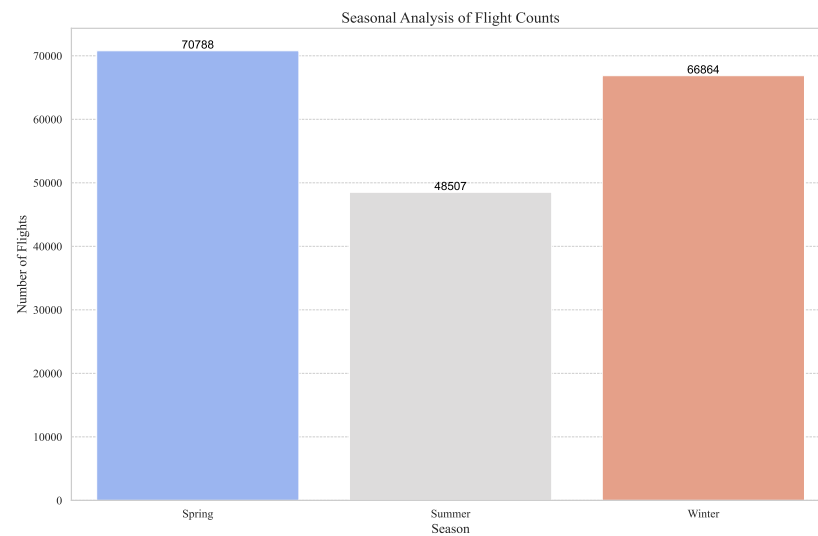


Figure 5. Seasonal trend analysis in 2023.

4.3.4. Weekend vs. Weekday Flight Analysis

The analysis of flight data reveals distinct differences between weekday and weekend operations, as shown in Figure 6. On weekdays, there were a total of 136,105 flights with an average arrival delay of approximately 9.59 min. In contrast, weekends recorded 50,054 flights with a slightly higher average arrival delay of 11.80 min. These findings suggest that while weekday travel is significantly more prevalent, weekend flights tend to experience longer delays on average, indicating potential operational challenges during peak leisure travel times.

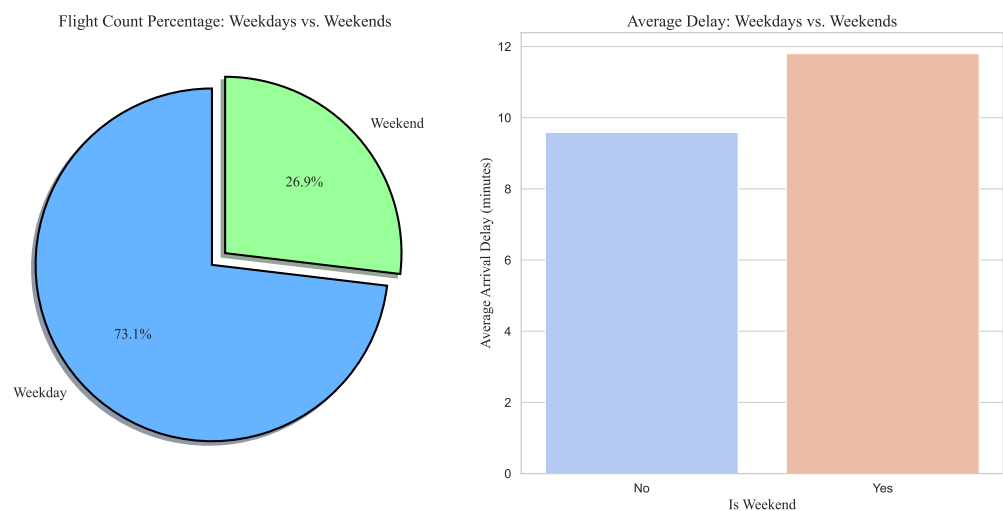


Figure 6. Weekday vs. weekend trend analysis.

4.3.5. Flight Count Analysis by Hour

As seen in Figure 7, the flight count data show clear trends for different times of day and week. From midnight to 4 a.m., flight activity is generally low, although Friday and Monday exhibit slightly higher counts, likely driven by business travel. Notably, Sundays show more activity than Saturdays, indicating the return of weekend travelers. As the clock strikes 5 a.m., flight counts increase dramatically, particularly during the early morning rush from 5 a.m. to 9 a.m., with Monday, Thursday, and Friday leading in numbers. This period is critical for business commuters, while weekends see reduced activity. However, Sundays still outperform Saturdays in morning flights, reflecting the conclusion of weekend trips. In the mid-morning hours (9 a.m. to 12 p.m.), flight activity remains robust but begins to taper after peak hours. Fridays continue to demonstrate high activity levels, likely due to a mix of business and leisure travel. Again, Sunday surpasses Saturday in flight counts, highlighting the significance of weekend travel. The afternoon (12 p.m. to 5 p.m.) experiences another surge in flights, especially on Mondays and Fridays, signaling ongoing business travel. Weekend activity is lower, but Sundays witness a rise as the day progresses. In the late afternoon (5 p.m. to 9 p.m.), another peak occurs, particularly on Monday and Friday evenings, while Sunday evenings see a sharp increase, reflecting the return of weekend travelers. Late-night hours (9 p.m. to midnight) show a significant drop in flight activity; however, Sunday nights remain busier than Saturday, as travelers conclude their weekend journeys. Overall, weekdays—especially Monday, Thursday, and Friday—dominate flight activity, with notable peaks in the morning and evening. Weekend travel is concentrated on Sundays, particularly in the morning and evening, as travelers prepare for the workweek.

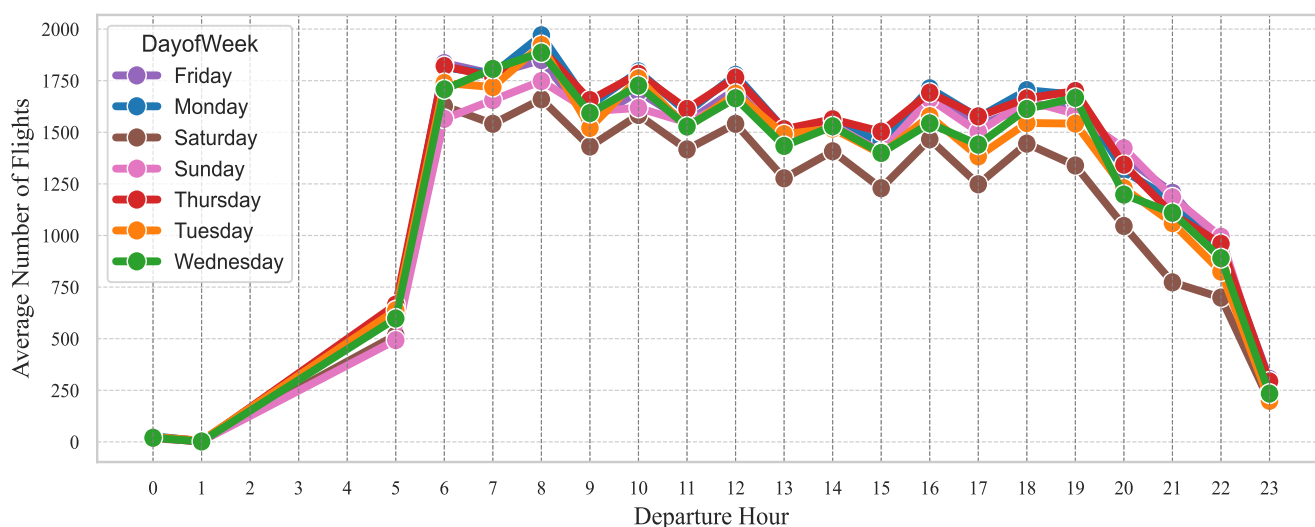


Figure 7. Average hourly flight frequency analysis.

4.3.6. Day vs. Night Flights Analysis

This study highlights (Figure 8) a striking disparity in flight delays between night and day flights. Night flights, representing only 8.46% of total operations (15,758 flights), suffer from an average delay of 18.68 min—over double the 9.40-min average delay of day flights, which comprise a substantial 91.54% (170,401 flights). This significant difference underscores a critical concern for travelers: while night flights may be fewer in number, they are far more likely to disrupt travel plans. Such insights are essential for informed scheduling and planning.

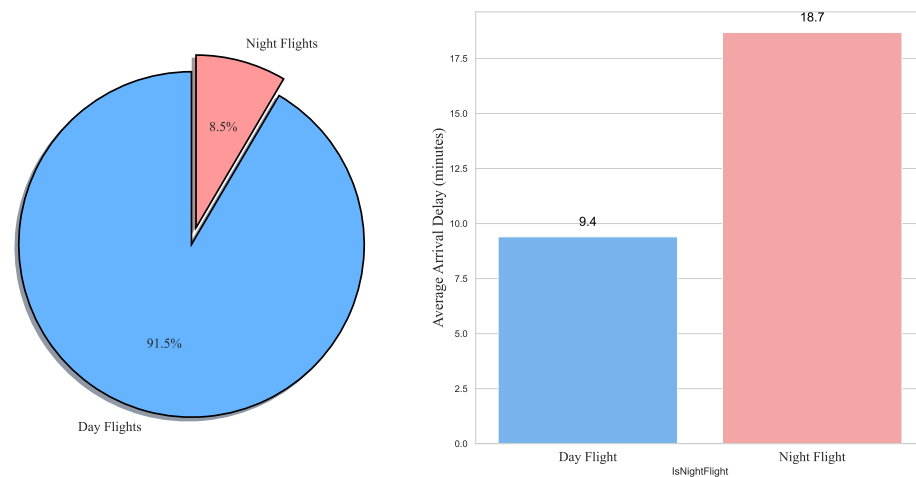


Figure 8. Day and night flight frequency and arrival delay analysis.

4.3.7. Extracted Time Related Data Analysis

The associations between time-based attributes and arrival delay (ARR_DELAY) (Figure 9) are typically minimal. The month, day of the month, day of the week, week of the year, and quarter all have correlations of around 0.03, indicating that these factors have no meaningful impact on arrival delays. Similarly, the departure time has a modest correlation of 0.12, showing a minor impact on delays. The departure minute, weekend status, and night flight status all have very poor associations with ARR_DELAY (0.02 to 0.04), indicating that these features have little impact on delays. Overall, time-based variables do not appear to be good predictors of arrival delays in this dataset.

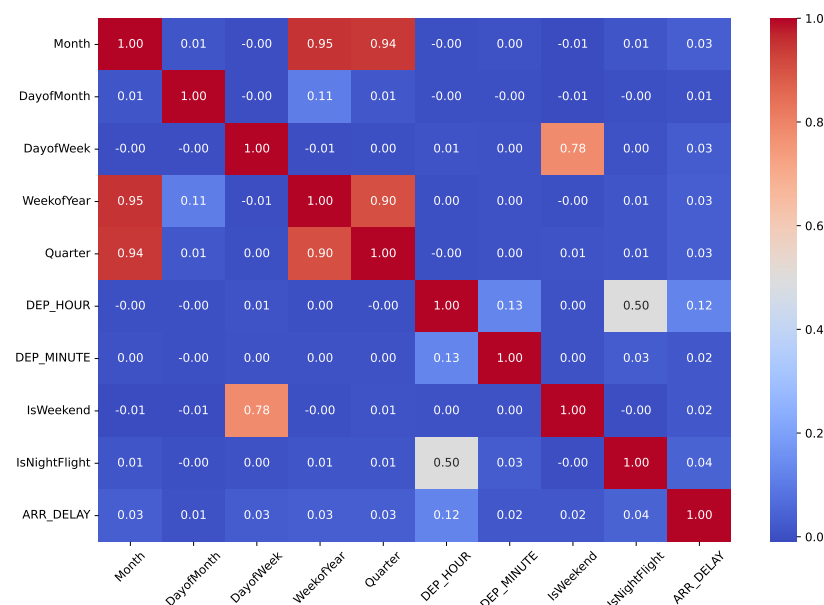


Figure 9. Correlation between time-based features and flight delays.

4.3.8. Previous Delay Related Feature Analysis

The empirical motivation behind the selection of these features lies in the understanding that previous delays, both departure and arrival, can influence current flight delays. Figure 10 shows that the moderate correlation (0.5) and clearer pattern in the scatter plot for Rolling_DEP_DELAY strongly suggest that it captures meaningful information regarding arrival delays and thus justifies its inclusion. Although the other features show weak correlations with ARR_DELAY, the scatter plots reveal that they still have some form of relationship with arrival delays, even if that relationship is not linear.

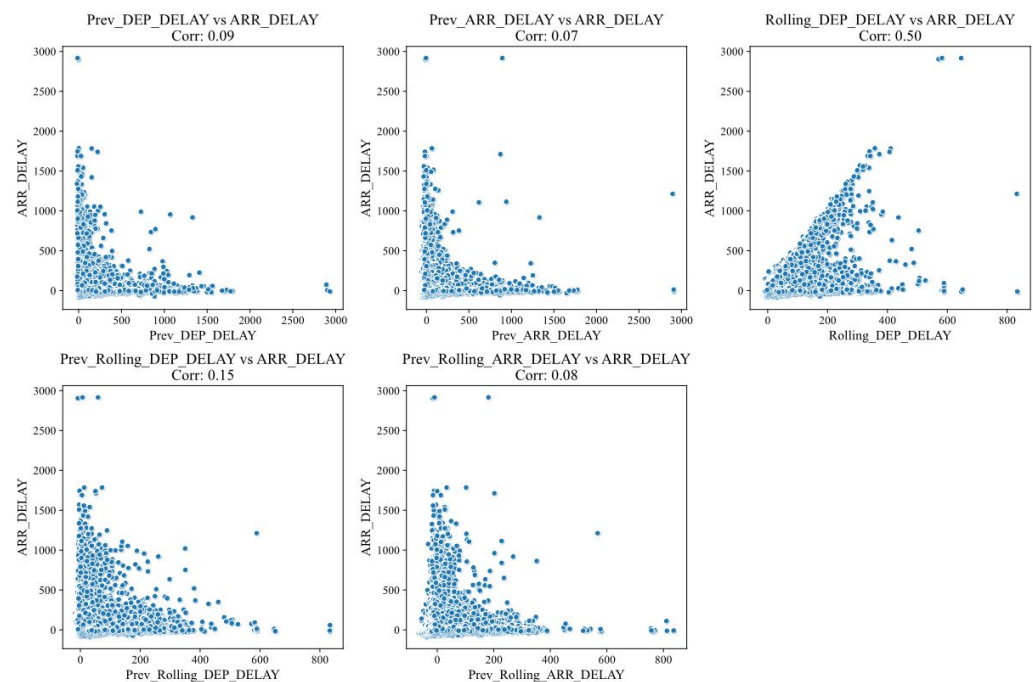


Figure 10. Scatter plots of previous delay features vs. ARR_DELAY.

4.3.9. Extracted Spatial Features

The correlation study shows (Figure 11) that, in spite of the paper's discussion of geographical factors such as clustering and haversine distance, these features barely affect arrival delays. In particular, there is very little relationship between haversine distance (0.01) and ARR_DELAY, and there are also weak correlations between ORIGIN_cluster (0.02) and DEST_cluster (0.01). These findings imply that delays are not much impacted by spatial patterns, such as the distance between airports and their geographic clusters.

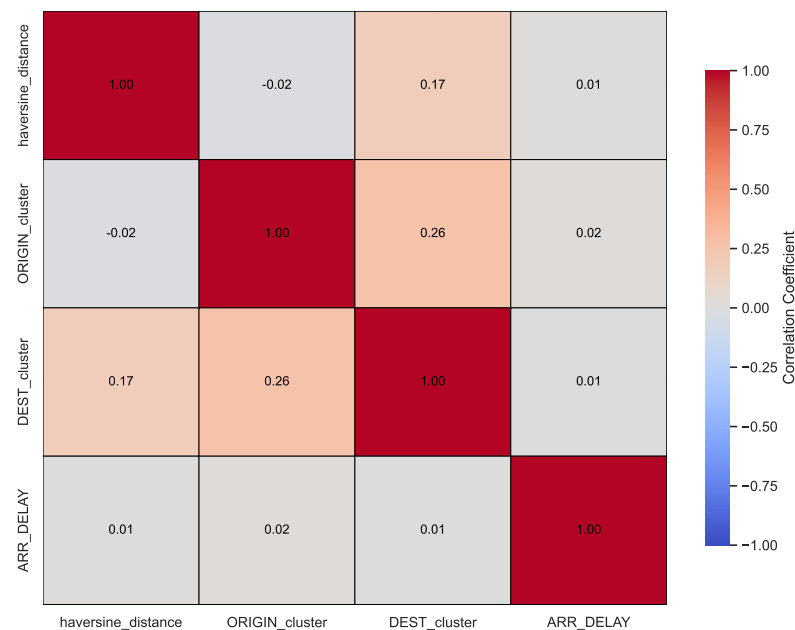


Figure 11. Heatmap of arrival delays with respect to haversine distance and clustering origin & destination.

4.3.10. Interactive Feature Correlation

The correlations for TAXI_DEP_DELAY, DISTANCE_DEP_DELAY, and DISTANCE_ARR_DELAY with ARR_DELAY are all strong (ranging from 0.80 to 0.84), indicating that these variables are relevant in predicting arrival delays. The regression lines (Figure 12) for these characteristics show obvious rising trends, supporting a concept that departure delays (both taxiing and distance-related) have a direct and considerable impact on overall arrival time.

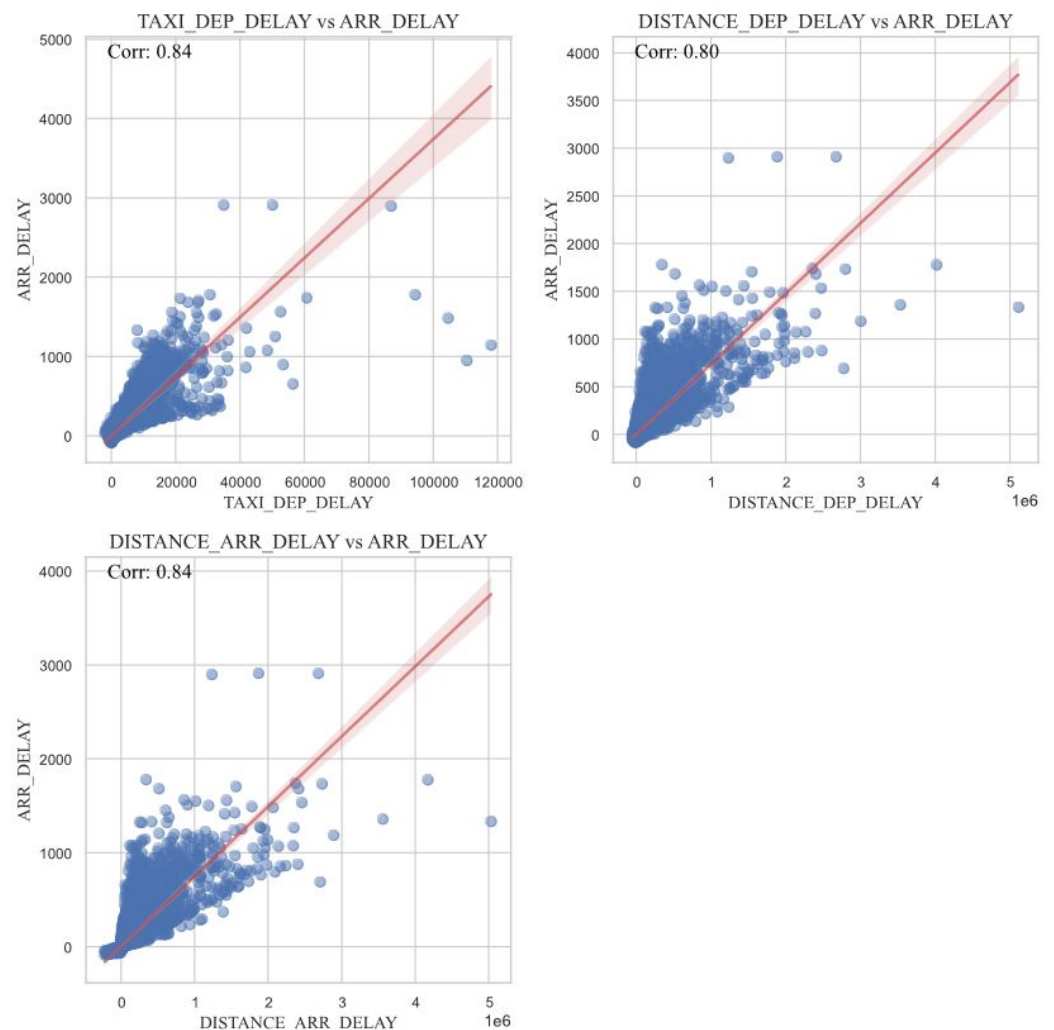


Figure 12. Regression Plots of interactive features vs. ARR_DELAY with correlation.

4.3.11. Additional Features Analysis

The correlation analysis shows (Figure 13) that the features IsNightFlight, sin_DEP_HOUR, cos_DEP_HOUR, Airline_Delay_Ratio_DEP, and Airline_Delay_Ratio_Prev_DEP have poor correlations with ARR_DELAY. Specifically, IsNightFlight shows a correlation of 0.04, showing that night flights have no meaningful effect on arrival delays. Similarly, sin_DEP_HOUR (−0.12) and cos_DEP_HOUR (0.06) show minimal correlations, indicating that the time of day, as reflected by these cyclical aspects, has little impact on arrival delays. The airline delay ratios, Airline_Delay_Ratio_DEP (0.03) and Airline_Delay_Ratio_Prev_DEP (0.03), show weak positive correlations, implying that the fraction of delays attributable to the airline, whether current or past, is not strongly associated with arrival delays.

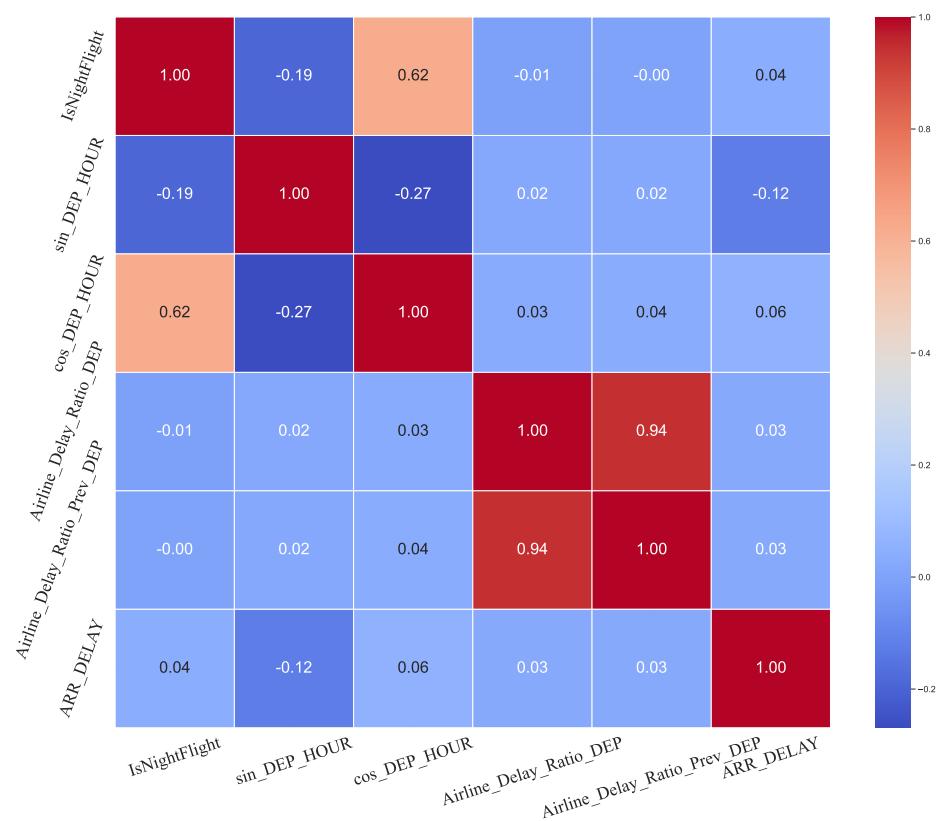


Figure 13. Correlation heatmap: additional features vs. ARR_DELAY.

4.4. Model Performance Comparison

The proposed DFFRN model demonstrates a significant improvement over other models in both training and testing performance. With a test RMSE of 1.672 and a near-perfect R^2 score of 99.916%, DFFRN consistently outperforms alternatives, showcasing superior accuracy and efficiency. Compared to LSTM, which achieves a test RMSE of 2.057, and CatBoost with a test RMSE of 3.884, the proposed model provides a marked reduction in error. Traditional models like HGBR and Linear Regression (LR) demonstrate reasonable accuracy but fail to reach the precision level achieved by DFFRN. This performance boost by the proposed model makes it an ideal choice for accurate, reliable predictions in this application.

The performance of several models is summarized in Table 7 according to training and testing metrics, with particular attention to RMSE, MAE, and R^2 values.

Table 7. Model performance metrics with proposed model highlighted.

Model Name	Train Results			Test Results			Run Time
	RMSE	MAE	R^2 %	RMSE	MAE	R^2 %	
HGBR	8.845	3.888	97.976	7.923	3.886	98.118	1.5 s
LR	6.891	4.709	98.772	6.661	4.645	98.670	0.66 s
DecisionTree	6.891	4.709	98.772	6.661	4.645	98.670	4.84 s
GBR	6.462	4.224	98.821	6.272	4.203	98.821	1 m 35 s
LGBMR	6.782	2.573	98.809	4.643	2.559	99.354	1.26 s
CNN	4.526	3.109	99.470	4.519	3.168	99.388	23 m 45 s
XGBoost	3.244	1.859	99.728	5.811	2.179	98.988	1.20 s
MLP	3.615	2.325	99.662	3.504	2.309	99.632	3 m 7 s
CatBoost	2.748	1.576	99.805	3.884	1.683	99.548	23 s
RF	1.895	0.546	99.907	3.892	1.371	99.546	6 m 1 s
LSTM	2.081	1.215	99.888	2.057	1.255	99.873	9 m 6 s
DFFRN (Proposed)	1.602	1.017	99.933	1.672	1.049	99.916	4 m 40 s

4.5. Predictive Accuracy Analysis of the Proposed Model

To assess our suggested model's prediction accuracy, we performed an analysis comparing the actual and predicted values through visualization. Figure 14 provides a two-part visual analysis, showcasing both a scatter plot for actual vs. predicted values and a line plot of their moving averages. The scatter plot on the left illustrates a reliable linear correlation between anticipated and real values, indicating a high R^2 score and demonstrating the model's precision. The ideal fit line ($y = x$) aligns closely with the data points, underscoring the model's accurate predictions. On the right, the line plot of moving averages further supports the model's consistency, with the predicted moving average closely tracking the actual values. This visualization highlights the model's effectiveness in producing reliable predictions across various data points.

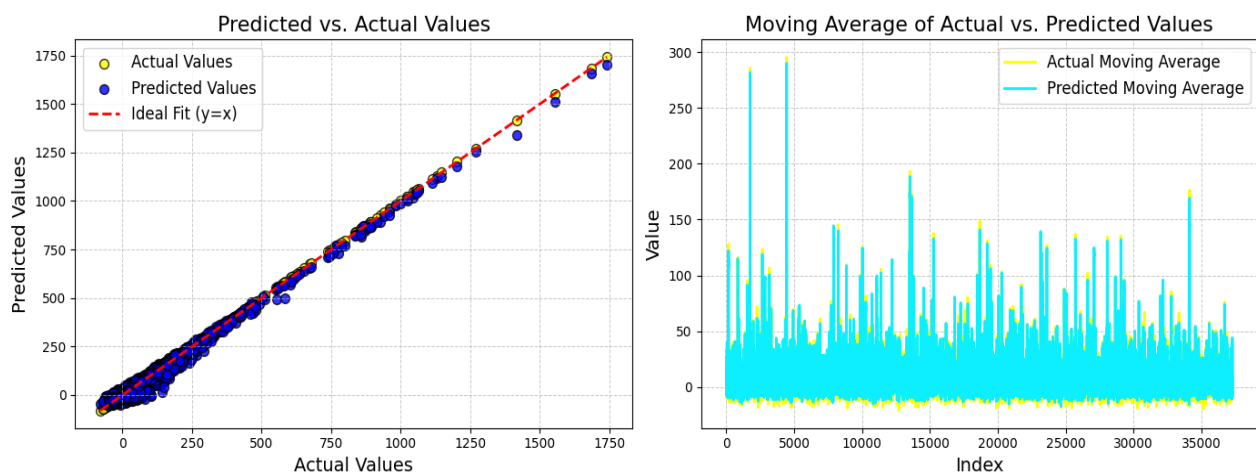


Figure 14. Visual analysis of actual vs. predicted values and moving averages for the proposed model.

4.6. Ablation Study

4.6.1. Before and After Feature Engineering Result Analysis

The ablation study in Table 8 quantifies the substantial performance gains achieved at each stage of our feature refinement process, demonstrating the power of our proposed correlation-based feature selection algorithm. Starting without feature engineering, the model displayed high error levels, with a training RMSE of 10.789 and a test RMSE of 10.716, yielding R^2 scores of 96.988% (train) and 96.558% (test). Upon applying feature engineering to refine the feature set to 50, we observed an impressive reduction in error: RMSE dropped by approximately 70% to 3.278 (train) and 3.025 (test), while R^2 scores surged to 99.722% and 99.726%, respectively. This significant improvement shows the foundational role of feature engineering in enhancing model accuracy, reducing overfitting, and stabilizing performance across datasets.

Table 8. Study of ablation: performance before and after feature engineering and feature selection.

Scenario	Train			Test		
	RMSE	MAE	R^2 %	RMSE	MAE	R^2 %
Before Feature Engineering	10.789	7.505	96.988	10.716	7.537	96.558
After Feature Engineering (50 features)	3.278	1.619	99.722	3.025	1.611	99.726

4.6.2. CFS Algorithm's Sensitivity Checking with Threshold Value

The sensitivity analysis of the CFS thresholds demonstrates (Table 9) how changing the threshold impacts both the amount of selected features and the model's performance. As the CFS threshold rises, the number of picked characteristics falls. For example, at a

threshold of 0.0005, 42 features were chosen; however, at a threshold of 0.05, just 19 were retained. The model worked optimally at a CFS threshold of 0.005, selecting 41 features with a test RMSE of 1.595, MAE of 0.786, and test R^2 of 99.924. As the threshold value increased, the model's performance marginally decreased, with test RMSE and MAE increasing, indicating that lowering the number of features beyond a certain point can affect predicted accuracy.

Table 9. Sensitivity analysis of CFS thresholds and model performance. Note: The values represented in bold format indicate better performance results.

Threshold	Features Selected	Train Results			Test Results		
		RMSE	MAE	R^2 %	RMSE	MAE	R^2 %
0.0005	42	1.601	1.017	99.934	1.672	1.048	99.916
0.005	41	1.414	0.762	99.948	1.595	0.786	99.924
0.01	36	1.602	1.017	99.933	1.672	1.049	99.916
0.02	29	1.883	1.201	99.908	1.991	1.245	99.881
0.05	19	2.131	1.254	99.883	2.194	1.274	99.856

5. Discussion

5.1. Implications for Passengers and Policymakers

While this study emphasizes the operational benefits for airlines and airports, the broader implications for passengers and policymakers are equally significant and merit detailed exploration. Incorporating these perspectives highlights the model's utility in fostering a more resilient and passenger-friendly aviation ecosystem while supporting evidence-based policy interventions.

1. Benefits for Passengers:

For passengers, accurate flight delay predictions translate into better planning and reduced stress. Airlines and airports can proactively communicate potential delays through notifications, enabling passengers to adjust travel plans, avoid unnecessary waiting times, and reduce missed connections. For example, a real-time implementation of this model could notify passengers of expected delays upon booking or check-in, empowering them with actionable information to optimize their travel experience. This improves overall satisfaction and loyalty, especially for frequent travelers. Moreover, the model can guide airlines to allocate resources efficiently, such as rebooking passengers on alternative flights or coordinating timely ground services. By minimizing disruptions, passengers are more likely to have a seamless journey despite operational delays. This is particularly crucial during peak seasons or adverse weather conditions when managing passenger expectations becomes challenging.

2. Guidance for Policymakers:

For policymakers, the model provides a robust analytical framework for identifying systemic issues contributing to flight delays, such as air traffic congestion, inefficiencies in airport operations, or resource allocation disparities. Regulatory bodies can leverage these insights to design targeted interventions, such as revising slot allocations, mandating minimum standards for delay communication, or incentivizing punctuality through policy mechanisms. For instance, the model's spatio-temporal insights can reveal delay patterns tied to specific routes, airports, or times of the year, allowing policymakers to address bottlenecks or introduce contingency plans in high-risk areas. Additionally, integrating this model into national or regional transportation planning systems can improve coordination between aviation and other modes of transport, such as rail or public transit, ensuring smoother passenger transitions during delays.

3. Strengthening Communication and Transparency:

By fostering collaboration between airlines, airports, and regulatory bodies, the model can establish a standardized protocol for sharing delay predictions. This ensures timely dissemination of accurate information to passengers, enhancing transparency and trust. Such communication could be institutionalized through mobile applications, airport displays, or automated messages, ensuring that passengers remain informed throughout their journey. Therefore, extending the benefits of this model to passengers and policymakers enhances its utility beyond operational efficiency, contributing to a more equitable, transparent, and passenger-centric aviation ecosystem. By integrating predictive capabilities into broader regulatory and communication frameworks, this model can play a pivotal role in shaping a resilient and responsive aviation industry.

5.2. Addressing Class Imbalances and Delay Distribution in the Model

In the current study, the delays were treated as continuous variables for regression purposes, with the goal of predicting the magnitude of delays. While the data cleaning process focused on excluding canceled and diverted flights (to ensure that we were working with operational data), we did not specifically assess or address whether the distribution of delays was normal or skewed. Typically, flight delay data tend to exhibit a right-skewed distribution, with a higher frequency of shorter delays and fewer instances of longer delays. This natural skewness could affect how well the model predicts more extreme delays, which are less frequent but highly impactful.

While class imbalances are more often associated with classification tasks, where one class is much more frequent than another, regression models also face challenges when predicting rare but significant outcomes. In this case, predicting long delays could be treated as a form of imbalance, as the model might overfit to more frequent short delays and fail to predict long delays accurately.

To address this, in future iterations of the study, we plan to include discussions and techniques to handle class imbalances and skewed distributions. Several strategies could be considered to improve the model's performance in such cases: One approach would be to use resampling methods, such as oversampling the rare long delays or undersampling the more common short delays. This would help balance the dataset and ensure that the model learns to predict long delays more effectively. Alternatively, synthetic data generation methods like SMOTE (Synthetic Minority Over-sampling Technique) could be explored to generate artificial long delay instances, preserving the overall dataset's integrity while improving model sensitivity to rare events. Another strategy to address the imbalance would be to apply weights to the loss function during model training. For example, higher penalties could be assigned to errors related to long delays, ensuring that the model places more importance on accurately predicting rare, significant delays. This approach would help to make the model more sensitive to the minority class of long delays, even when they are less frequent.

5.3. Addressing Ethical Concerns and Data Privacy in Flight Delay Prediction

The ethical considerations surrounding the use of sensitive operational data in flight delay prediction models are critical, especially in the context of privacy, fairness, and the potential misuse of predictive tools. The proposed model relies on flight details that could include sensitive operational information, such as departure schedules, delays, and logistical patterns. Therefore, ensuring robust data privacy measures and mitigating risks of unintended consequences is of paramount importance.

To safeguard data privacy, all operational data used in this study were anonymized prior to analysis, with identifying details such as specific flight numbers, airline personnel data, and passenger information excluded. Anonymization was coupled with strict access controls to ensure that data handling complied with prevailing data protection regulations, such as the General Data Protection Regulation (GDPR) or equivalent national standards.

These measures ensure that while the data's analytical value is preserved, individual privacy and sensitive operational details remain secure.

Furthermore, to address concerns of potential misuse, such as scheduling biases, the predictive model was developed with an emphasis on transparency and fairness. For example, care was taken to avoid embedding features or patterns that could unintentionally favor specific airlines, routes, or time slots, which might exacerbate systemic inequalities in scheduling. The feature engineering process deliberately focuses on spatio-temporal factors and operational metrics that are generalizable and devoid of discriminatory biases.

Additionally, we acknowledge the broader ethical implications of predictive modeling in aviation, particularly the risk of introducing unintended biases or influencing resource allocation in a way that might disproportionately impact smaller airlines or less-frequented routes. To counter this, future iterations of this study could include an audit framework to systematically assess the fairness and equity of model recommendations across different stakeholders. Incorporating stakeholder feedback, including from passengers and industry professionals, can further ensure ethical deployment.

Therefore, while the proposed model offers significant utility in improving operational efficiency and passenger satisfaction, ethical vigilance remains essential. By adhering to strict data privacy practices, ensuring model fairness, and engaging in continuous ethical evaluation, this study strives to contribute responsibly to the data-driven transformation of the aviation sector.

5.4. Limitation of the DFFRN Model

The proposed model exhibits significant overfitting when trained on a small dataset, as seen in Figure 15. The training MAE steadily drops, whereas the validation MAE rises after a certain point, indicating that the model is fitting too closely to the training data and learning noise rather than generalizable patterns. This leads to poor performance on the validation set, highlighting the overfitting issue.

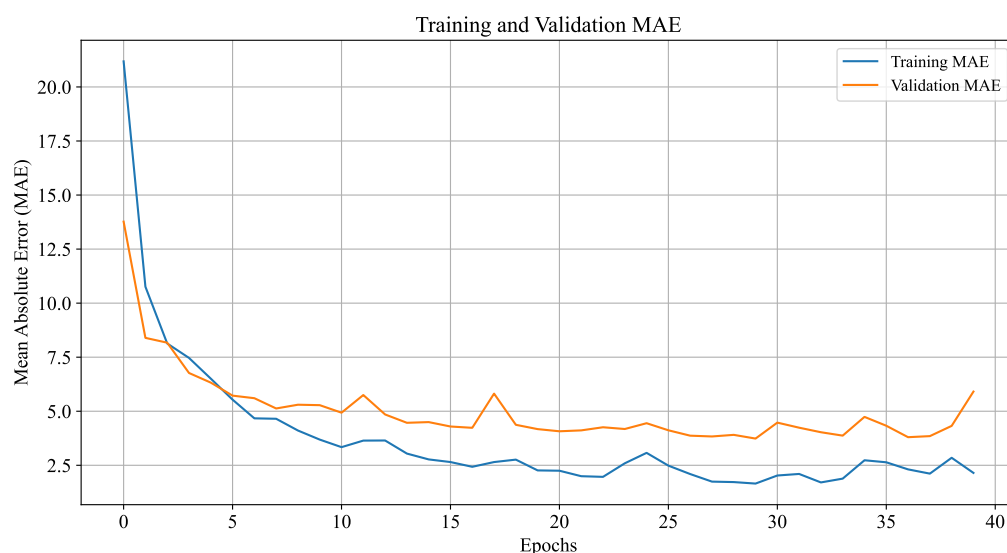


Figure 15. Comparison of training and validation MAE over epochs with small dataset before applying regularization and dropout.

To solve this issue, regularization approaches such as L2 regularization and dropout were used during training. L2 regularization penalizes high weights, whereas dropout keeps the model from becoming unduly reliant on certain characteristics. These changes resulted in a more stable training–validation curve (Figure 16), with the validation MAE no longer growing after a given epoch, indicating enhanced generalization and performance.

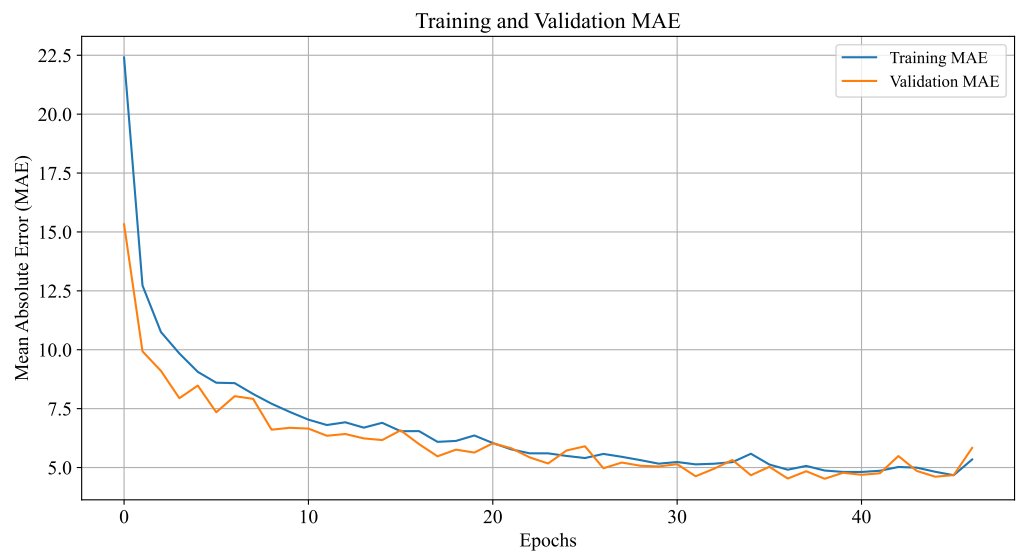


Figure 16. Comparison of training and validation MAE over epochs with small dataset after applying regularization and dropout.

6. Conclusions

This study presented a comprehensive approach to flight delay prediction, starting with data cleaning and preprocessing to ensure data quality and integrity. After refining the dataset, we proceeded with Exploratory Data Analysis (EDA) to uncover key patterns and trends impacting flight delays. This phase involved analyzing delays across different states, exploring the relationship between departure and arrival delays, and examining seasonal, hourly, and day–night trends. These insights established a foundational understanding of delay patterns, which informed our subsequent feature engineering and selection strategies. We applied advanced spatio-temporal feature engineering to enrich the dataset, capturing crucial spatial and temporal dependencies in flight delay dynamics. This enabled the model to account for regional and time-based variations, further enhancing predictive accuracy. To streamline the input data and improve efficiency, we proposed a novel Correlation-Based Feature Selection (CFS) algorithm. This algorithm effectively reduced the feature set from 50 to 36, achieving an approximately 50% reduction in RMSE while maintaining high accuracy. The CFS algorithm’s ability to retain only the most predictive features not only boosted model performance but also optimized computational resources, underscoring its practical value. Among the various models tested, our proposed DFFRN model stood out, achieving superior performance metrics, including a test RMSE of 1.672 and an R^2 score of 99.916%. These results demonstrate DFFRN’s robustness and precision in predicting flight delays, particularly when paired with our spatio-temporal feature engineering and CFS algorithm. This combination significantly enhanced predictive power, making DFFRN a reliable tool for delay prediction in complex aviation contexts. Looking ahead, we plan to expand this framework by developing a real-time flight delay monitoring system, allowing for immediate, data-driven insights for stakeholders. We will also explore more sophisticated feature selection algorithms to adapt dynamically to changing conditions, further enhancing predictive accuracy and applicability. This research lays the groundwork for integrating predictive analytics into aviation management, offering a powerful approach to flight delay forecasting that balances accuracy, efficiency, and practical relevance for real-world applications. Moreover, we also plan to incorporate weather-related features by integrating external meteorological datasets to enrich the input data and address the current limitation of missing weather-related variables. This will allow us to evaluate the impact of adverse weather conditions on flight delays more comprehensively.

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